Type II Models for Degeneration of K3 Surfaces

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Abstract

This thesis develops a uniform, combinatorial framework for Type II degenerations of polarized K3 surfaces, building on Alexeev-Brunyate-Engel's model [ABE22] of integral-affine spheres (IAS^2) for Type III degeneration of ellitic K3 surfaces. The construction reinterprets geometric invariants by simple polygonal measurements, and scales without case-by-case analysis as the degree grows, thereby placing Type II and Type III degenerations in a single integral-affine/tropical framework tightly linked to mirror symmetry and the Gross-Siebert program.

Contents

1. Introduction 1
2. Preliminaries 3
2.1. Kulikov models and anticanonical pairs 3
2.2. Topology of Kulikov models 6
3. Type III degeneration via integral affine sphere 7
3.1. Integral-affine surfaces and pseudo-fans 7
3.2. IAS^2 and Type III Kulikov Surfaces 8
3.3. Modifications and base change 10
3.4. IAS^2 via symplectic geometry and the monodromy theorem 11
3.5. Example: Degeneration of elliptic K3 surfaces 13
4. Type II degeneration 17
4.1. Converting to Type III degeneration of ellipitic K3 surfaces 17
4.2. Type II Kulikov surface via lattice polygon 19
4.2.1. Construction of polarized Type II anticanonical pairs 19
4.2.2. Construction of d-semistable Kulikov surfaces 20
4.2.3. Construction of polarized Kulikov models 22
4.3. Examples 26
4.4. Generalizations and Conjectures 28
References 30

1. Introduction

The study of degenerations of K3 surfaces plays a central role in understanding the geometry at the boundary of moduli spaces. By the Torelli theorem, the coarse moduli space F_{2d} of principally polarized K3 surfaces (X, L) with polarization degree $L^2 = 2d$ can be realized as a quotient of a 19-dimensional Hermitian symmetric domain by an arithmetic group. This space admits both the Baily–Borel compactification F_{2d}^{BB} and infinitely many (semi-)toroidal compactifications F_{2d}^{tor} and F_{2d}^{semi} [Loo03]. A natural problem is to describe the geometric objects corresponding to the boundary points of these compactifications.

Building upon earlier work of Kulikov [Kul77] and the Picard–Lefschetz theory, Engel proposed a method to reconstruct the central fiber of a Type III Kulikov degeneration of K3 surfaces directly

from a given monodromy transformation. In subsequent works [Eng18, ABE22, AET23], the combinatorial type of a Kulikov model is described via the dual complex, a triangulated two-sphere, endowed with an integral-affine structure. The 24 integral-affine singularities encode the geometry through *pseudo-fans*, while symplectic methods allow one to compute the monodromy invariant of a degeneration, as the class of a symplectic form of a Lagrangian fibration over the dual complex of the central fiber. This fits naturally into the framework of the Gross–Siebert program, which seeks to understand mirror symmetry near maximally unipotent degenerations of Calabi–Yau varieties using tropical and integral-affine geometry.

In certain (semi-)toroidal compactifications determined by a fan in the rational closure of the positive cone, the degeneration type may change as the monodromy invariant approaches faces of the fundamental chamber. In particular, when the associated integral-affine sphere degenerates to a segment, one obtains a Type II degeneration. While explicit constructions of Type II Kulikov models are known in low degree—e.g. the types E_8A_1, $D_{10}E_7$, $D_{16}A_1$, and A_{17} for degree 2 K3 surfaces in [Fri84], and a recent work [Jon25] for degree 4 K3s—the number of distinct types grows quickly with the degree [Sca87], making case-by-case approaches infeasible.

This motivates the search for a combinatorial framework for Type II degenerations, analogous to the integral-affine sphere description for Type III. Recall that a Type II Kulikov model is an snc chain of surfaces $X_0 = V_0 \cup \cdots \cup V_n$ with smooth elliptic double loci $D_{i,i+1} := V_i \cap V_{i+1}$ and no triple points. The guiding question of this work is therefore:

> **Question 1.1.** Can one define a suitable combinatorial structure on a segment that allows the reconstruction of a Type II Kulikov model from monodromy data?

We present the construction in this thesis, using the model in [ABE22]. Our approach proceeds in two complementary steps.

Step 1: Relating Type II to Type III via lattice theory. If $h = c_1(L) \in H^2(X, \mathbb{Z})$ is the class of the primitive polarization, the Type II boundary strata are uniquely determined by Γ-orbits of primitive rank-2 isotropic lattices $I \subset h^\perp$ where $\Gamma := \{\gamma \in \Lambda_{\mathrm{K3}} : \gamma(h) = h\}$. Thus we define the monodromy data as a *Type II pair* (I, h), consisting of a primitive rank-2 isotropic sublattice $I \subset \Lambda_{\mathrm{K3}}$ and a primitive polarization class h with $h^2 = 2d$ and $(I, h) = 0$. Using the unimodularity of Λ_{K3}, we identify the orthogonal quotient $I^\perp/I \cong \mathrm{II}_{1,17}$. This yields a natural map from $O(\Lambda_{\mathrm{K3}})$-orbits of Type II pairs to $O(\mathrm{II}_{1,17})$-orbits of vectors $\lambda \in \mathrm{II}_{1,17}$ with $\lambda^2 = 2d$. Conversely, Eichler transformations control lifts of v back to Type II data, leading to a bijection:

$$\{O(\Lambda_{\mathrm{K3}})\text{-orbits of Type II pairs}\} \longleftrightarrow \{O(\mathrm{II}_{1,17})\text{-orbits of vectors of positive norm}\}.$$

The right-hand data, orbits of positive $\lambda \in \mathrm{II}_{1,17}$, is the *monodromy invariant* of the Type III degeneration of elliptic K3s. As a corollary, monodromy types of Type II degenerations of primitively polarized degree-$2d$ K3 surfaces correspond to monodromy types of Type III degenerations of elliptic K3 surfaces with $2d$ triple points in the central fiber by [FS86].

Step 2: A combinatorial model for Type II via lattice polygons. For $\lambda \in \mathrm{II}_{1,17}$ in the fundamental chamber of $O^+(\mathrm{II}_{1,17})$, the work of [ABE22] assigns a triangulated integral-affine sphere $B(\lambda)$ (an IAS^2) and constructs a Type III Kulikov surface $X_0(\lambda)$, together with a canonical polarizing divisor

compatible with an elliptic fibration. From the underlying lattice polygon $Q(\lambda)$ of $B(\lambda)$, we introduce a segment model for Type II by cutting $Q(\lambda)$ into vertical strips (trapezoids or end-triangles) determined by the bending points of the associated PL function. To each strip we associate an explicit polarized Type II anticanonical pair $(V_i, D_i; h_i)$, chosen from a finite menu of models (blow-ups of $\mathbb{P}^2$, $\mathbb{P}^1 \times \mathbb{P}^1$, or ruled surfaces over an elliptic curve with prescribed sections), so that:

- the integral-affine area of the strip equals h_i^2, and the strip's boundary lengths record the intersection numbers $h_i \cdot D_{i,i-1}$ and $h_i \cdot D_{i,i+1}$;
- adjacent strips glue along elliptic boundaries (after some blow-ups) to produce a chain $X_0 = \cup V_i$ with double curves $D_{i,i+1} \cong E$, satisfying the Type II Kulikov constraint $D_{i,i+1}^2 + D_{i+1,i}^2 = 0$;

We then vary the gluing translations and the blow-up centers to hit the d-semistable locus [FM83a], thereby obtaining a d-semistable Type II Kulikov surface $X_0(h)$ that smooths to a polarized K3 family $X(h) \to \Delta$. Numerically Cartier compatibility along the double locus gives a global limit polarization on the Kulikov model.

Consequences. The construction provides a uniform combinatorial recipe to pass from monodromy data in $\mathrm{II}_{1,17}$ to polarized Type II Kulikov models, avoiding case-by-case analysis even when the degree is large. It also places Type II and Type III degenerations into a single integral-affine/tropical framework: Type III is governed by triangulated IAS^2 on S^2; Type II arises from their degeneration to a segment encoded by the lattice polygon $Q(\lambda)$ and its strip decomposition. Along the way, we recover and re-interpret known invariants (charges, period obstructions) in terms of simple polygonal measurements, and we exhibit explicit examples (including low-degree cases) illustrating the method.

Organization of the thesis. Section 2 recalls basic facts on Kulilkov degeneration of K3 surfaces. Section 3 reviews Engel's Type III degeneration via integral-affine geometry; we include the elliptic K3 case and the Coxeter chamber description in $\mathrm{II}_{1,17}$ following [ABE22]. Section 4 develops the Type II theory, where the constructions and proof of main theorems are given.

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2. Preliminaries

2.1. Kulikov models and anticanonical pairs.

We recall some well-known results about degeneration of K3 surfaces as preliminaries.

Theorem 2.1. *[Kul77] [PP81] Let $\mathcal{X} \to (C, 0)$ be a flat proper family over a germ of a curve such that the fibers of $\mathcal{X}^* \to C^* = C\backslash 0$ are projective K3 surfaces. Then there is a finite ramified base change $(C', 0) \to (C, 0)$ and a birational modification $\mathcal{X} \to \mathcal{X} \times_C C'$ such that $\pi : \mathcal{X}' \to C'$ is semistable (a smooth threefold with $\mathcal{X}'_0$ a reduced normal crossing divisor) with $\omega_{\mathcal{X}'/C'} \backsimeq \mathcal{O}_{\mathcal{X}'}$.*

We call such degeneration $\mathcal{X} \to (C, 0)$ a Kulikov degeneration and the central fiber $\mathcal{X}_0$ a Kulikov surface. Let $V_i \subset \mathcal{X}_0$ denote the irreducible components, $D_{ij} = V_i \cap V_j$ and $T_{ijk} = V_i \cap V_j \cap V_k$ the

double curves and triple points, respectively. By convention, we write $D_{ij} \subset V_i$ and $D_{ji} \subset V_j$. Let $D_i = \sum_j D_{ij}$ be the part of the double locus of $\mathcal{X}_0$ lying on V_i.

Let $N := \log T = (T - Id) - \frac{1}{2}(T - Id)^2$ be the logarithm of the unipotent Picard-Lefschetz transformation $T : H^2(\mathcal{X}_t, \mathbb{Z}) \to H^2(\mathcal{X}_t, \mathbb{Z})$. It is a nilpotent endomorphism satisfying $N^3 = 0$, hence determining a monodromy weight filtration $W_\bullet(N)$ on $H^2(\mathcal{X}_t, \mathbb{Q})$ defined by

$$\{0\} \subseteq W_0 = \mathrm{Im}\, N^2 \subseteq W_1 = \mathrm{Ker}\, N \cap \mathrm{Im}\, N \subseteq W_2 = \mathrm{Ker}\, N + \mathrm{Im}\, N \subseteq W_3 = \mathrm{Ker}\, N^2 \subseteq W_4 = H^2(\mathcal{X}_t, \mathbb{Q}).$$

N is actually an integral endomorphism and see [Sch73] for more generalities.

We have the following three types of degenerations given by this arithmetic and corresponding Kulikov models.

Proposition 2.2. *There are three possible cases for the order of N, called Types I, II, III:*

(I) If $N = 0$, then $\mathcal{X}_0$ is a smooth K3 surface.

(II) If $N^2 = 0$ but $N \neq 0$, then the filtration has the form $0 \subset W_1 \subset W_1^\perp = W_2 \subset H^2(\mathcal{X}_t, \mathbb{Q})$ and dim $W_1 = 2$. The Kulikov surface $\mathcal{X}_0 = \bigcup_{i=1}^n V_i$ is a chain of surfaces with dual complex a segment. The ends V_1 and V_n are rational and V_i for $i \neq 1, n$ are $\mathbb{P}^1$-bundles over E for a fixed elliptic curve E. The double curves $D_{i,i+1} := V_i \cap V_{i+1}$ are isomorphic to E, the union of the double curves lying on V_i is an anticanonical divisor, and

$$D_{i,i+1}^2 + D_{i+1,i}^2 = 0$$

.

(III) If $N^3 = 0$ but $N^2 \neq 0$, then the filtration has the form $0 \subset W_0 \subset W_0^\perp = W_2 \subset H^2(\mathcal{X}_t, \mathbb{Q})$ and dim $W_0 = 1$. The Kulikov surface $\mathcal{X}_0 = \bigcup_{i=1}^n V_i$ is a union of rational surfaces whose dual complex is a triangulation of the sphere. The union of all double curves lying on (the normalization of) V_i form an anticanonical cycle of rational curves. Declaring $d_{ij} := -2p_a(D_{ij}) - D_{ij}^2$, we have

$$d_{ij} + d_{ji} = -2. \tag{2.1}$$

Furthermore, $Nx = (x \cdot \lambda)\delta - (x \cdot \delta)\lambda$ for $\delta \in H^2(\mathcal{X}_t, \mathbb{Z})$ a primitive isotropic vector, and $\lambda \in \delta^\perp/\delta$ satisfying $\lambda^2 = \#\{$triple points of $\mathcal{X}_0\}$. When $\lambda^2 = 0$, the imprimitivity of λ is the number of double curves of $\mathcal{X}_0$.

The core of the series of theories, originating from [FS86], is that the arithmetic of N reflects the combinatorics of central fiber $\mathcal{X}_0$. The arithmetic of N is in turn characterized by two invariants. By the proposition above, the filtration is determined by a single isotropic subspace W_0 or W_1 respectively. Define I as the intersection of the isotropic subspace and the K3 lattice Λ_{K3}. Then $I = \mathbb{Z}\delta$ in Type III and $I = (\mathbb{Z}\delta \oplus \mathbb{Z}\lambda)^{sat}$ in Type II. We have

Proposition 2.3 (arithmetic invariants of the monodromy)**.** *Let $I_d(\Lambda_{K3})$ denote the set of primitive rank-d isotropic sublattices in Λ_{K3}. The endomorphism N, up to conjugation of $O(\Lambda_{K3})$, is determined by its index k (the positive integer such that $N = kN'$ with N' primitive) and*

Type II: the class of I in $I_2(\Lambda_{K3})/O(\Lambda_{K3})$;

Type III: the class of I in $I_1(\Lambda_{K3})/O(\Lambda_{K3})$ and the class of a vector $\lambda \in I^\perp/I$ with $(\lambda, \lambda) > 0$ module $O(I^\perp/I)$.

Definition 2.4. For Type III degenerations, we call $\lambda \in I^\perp/I$ the *monodromy invariant*.

Remark 2.5. In fact, rank 1 or 2 isotropic sublattices of Λ_{K3} are unique up to isometry. But in the polarized setting, one replaces the K3 lattice Λ_{K3} by the sublattice $h^\perp$. Then the uniqueness no longer holds.

Conversely there is the condition for a Kulikov surface to be smoothable:

Theorem 2.6. *Let X_0 be a Kulikov surface. Then X_0 deforms to a smooth K3 surface if and only if it is d-semistable, that is,*

$$\mathcal{E}xt^1(\Omega_{X_0}, \mathcal{O}_{X_0}) \cong \mathcal{O}_{(X_0)_{Sing}}.$$

To study combinatorics of Kulikov surfaces, we firstly recall some useful results about anticanonical pairs, following [FM83a].

Definition 2.7. An anticanonical pair (V, D) is a smooth rational surface V together with a cycle of smooth rational curves $D = D_1 + \cdots + D_n \in |-K_V|$. The *charge* of an anticanonical pair (V, D) is defined as $Q(V, D) := 12 - \sum(D_i^2 + 3)$.

In particular, we see that a toric pair (V, D) is an anticanonical pair in the sense that V is toric and D is the toric boundary.

Definition 2.8. A *corner blow-up* of (V, D) is the blow-up at a node of the cycle D and an *internal blow-up* is a blow-up at a smooth point of D. In both cases, the blow-up has an anticanonical cycle mapping to D. The corner blow-up leaves the charge invariant, while the internal blow-up increases the charge by 1.

More precisely, when $n > 1$ an internal blow-up on D_i increases the i-th *negative self-intersection* $d_i := -D_i^2$ by 1, and a corner blow-up at $D_i \cap D_{i+1}$ increases d_i and d_{i+1} by 1 and the new cycle has negative self-intersection 1. In addition to blow-ups on D, we can smooth any node of D:

Proposition 2.9. *Let (V, D) be an anticanonical pair and let p be a node of D. There exists a family of anticanonical pairs*

$$(\mathcal{V}, \mathcal{D}) \to \Delta$$

over the disc whose central fiber is (V, D) such that $\mathcal{D} \to \Delta$ is a smoothing of the node p.

Then a node smoothing at $D_i \cap D_{i+1}$ merges the two cycles into one with negative self-intersection $d_i + d_{i+1} - 2$ and increases the charge by 1.

Remark 2.10. By [Fri16] the pair (V, D) is toric, if and only if $Q(V, D) = 0$. Otherwise, $Q(V, D) > 0$. In particular, after some corner blow-ups, one can blow down any anticanonical pair to a toric pair (V, D) [GHK14].

The dual complex $\Gamma(\mathcal{X}_0)$ of Type III Kulikov surfaces plays an important role in the study of Kulikov degenerations. The following proposition in [FM83a] actually reflects the integral-affine structure on the dual complex $\Gamma(\mathcal{X}_0)$, as we will see in the next section.

Proposition 2.11 (Conservation of Charge)**.** *Let $\mathcal{X} \to (C, 0)$ be a Type III Kulikov degeneration. Then $\sum_{i=1}^n Q(V_i, \sum_j D_{ij}) = 24$. In particular, at most 24 components of X_0 are non-toric.*

2.2. **Topology of Kulikov models.** For a simple normal crossing degeneration $\mathcal{X} \to (C, 0)$, there is a deformation-retraction $c_t : \mathcal{X} \times [0, 1] \to \mathcal{X}_0$ called Clemens collapsing. So we have $H^*(\mathcal{X}, \mathbb{Z}) = H^*(\mathcal{X}_0, \mathbb{Z})$ and a map $c_t^* : H^2(\mathcal{X}_0, \mathbb{Z}) \to H^2(\mathcal{X}_t, \mathbb{Z})$.

The topology of Kulikov surfaces is studied in [FS86] via Mayer-Vietoris sequence. It follows that for Type III degeneration there is an exact sequence

$$0 \to \mathbb{Z} \to H^2(\mathcal{X}_0, \mathbb{Z}) \to \widetilde{\Lambda} \to 0.$$

Definition 2.12. The numerically Cartier classes on a Kulikov surface

$$\widetilde{\Lambda} = \widetilde{\Lambda}(\mathcal{X}_0) := \ker \left(\bigoplus_i H^2(V_i, \mathbb{Z}) \to \bigoplus_{i<j} H^2(D_{ij}, \mathbb{Z}) \right)$$

are collection of classes (α_i) such that $n_{ij} = \alpha_i.D_{ij} = \alpha_j.D_{ji}$.

Similarly for Type II degeneration one can deduce an analogous exact sequence $0 \to \mathbb{Z}^2 \to H^2(\mathcal{X}_0, \mathbb{Z}) \to \widetilde{\Lambda} \to 0$. We denote the left-hand terms $\mathbb{Z}$ or $\mathbb{Z}^2$ in the exact sequences by I_0. By Clemens-Schmid sequence the image $c_t^*(I_0)$ is always identified with $I \subset H^2(\mathcal{X}_t, \mathbb{Z})$.

Definition 2.13. Define the intersection form on $\widetilde{\Lambda}$ by $(\alpha_i).(\beta_i) = \sum_i \alpha_i.\beta_i$.

Definition 2.14. Define $\xi_i = \sum_j (D_{ji} - D_{ij}) \in \widetilde{\Lambda}$. It is easy to see that $\sum_i \xi_i = 0$ and $\xi_i.\widetilde{\Lambda} = 0$. Define $\Xi := \mathbb{Z}\text{-span}\{\xi_i\} \subset \widetilde{\Lambda}$ and declare $\Lambda := \widetilde{\Lambda}/\Xi$.

Proposition 2.15. *[AE23, 3.20] The map $c_t^* : H^2(\mathcal{X}_0) \to H^2(\mathcal{X}_t)$ induces a surjection $\widetilde{\Lambda} \twoheadrightarrow \{\delta, \lambda\}^\perp/I$ sending Ξ to zero, which thus descends to Λ. Furthermore, $\Xi = \widetilde{\Lambda}^{\text{null}}$ is the null sublattice. Hence Ξ is saturated, Λ is torsion-free and the induced map $\Lambda \to \{\delta, \lambda\}^\perp/I$ is an isometry of lattices.*

Proof. [FS86, 4.13] gives an exact sequence

$$0 \to \Xi \to \widetilde{\Lambda} \to \{\delta, \lambda\}^\perp/I \to 0.$$

Since the third term is torsion-free, the kernel Ξ must be the saturated. It is exactly the null lattice because the target $\{\delta, \lambda\}^\perp/I$ is nondegenerate and c_t^* preserves the intersection form. □

The definition 2.12 of numerically Cartier classes is independent of the d-semistability of the Kulikov surface X_0, so we can generalize it to a Kulikov surface not necessarily d-semistable.

The *period map* of X_0 is a homomorphism $\widetilde{\psi}$ from $\widetilde{\Lambda}(X_0)$ to $\mathbb{C}^*$ in Type III or the elliptic double curve E in Type II, which measures the obstruction to a class being represented by a Cartier divisor. The explicit construction for any Kulikov surface is given in [AE23, Construction 4.2, 4.3]. We use the notation loc.cit $\widetilde{\psi}_{X_0} \in \text{Hom}(\widetilde{\Lambda}, \mathbb{C}^*)$ or $\text{Hom}(\widetilde{\Lambda}, E)$ for the period maps.

Proposition 2.16. *[AE23, Prop 4.4] The surface X_0 is smoothable if and only if the period point $\widetilde{\psi}_{X_0}$ descends to a period point $\psi_{X_0} \in \text{Hom}(\Lambda, \mathbb{C}^*)$ or $\text{Hom}(\Lambda, E)$.*

Remark 2.17. From Prop 2.15 one sees that the period of a d-semistable Kulikov surface X_0 can be related to the monodromy of its smoothing $X \to (C, 0)$. There is some compatibility between the periods of d-semistable Kulikov surfaces and smooth K3 surfaces. In [AE23] the authors defined the *mixed marking* of a degeneration of *admissible* K3 surfaces and described the deformation spaces for d-semistable Kulikov surfaces.

3. Type III degeneration via integral affine sphere

This section serve as a review of Engel's series of theories on the degeneration of K3 surfaces and their applications. In particular, the construction in Section 3.5 is an inspiration point for the new construction of Type II models in the next section.

3.1. **Integral-affine surfaces and pseudo-fans.** One can study toric degeneration by tropical geometry, where the dual complex serve as an important combinatorial structure related to the toric fan [NS06] [Kat08]. Following this idea we can also study Type III Kulikov degeneration and anticanonical pairs by some generalization.

By Proposition 2.2, the dual complex of a Type III Kulikov surface is a triangulated sphere. Shape of the (convex) triangulations of spheres is studied by [Thu98]: with a locally Euclidean metric, any such triangulation is visualized as the gluing of some Euclidean triangles along vertex and edges in $\mathbb{R}^2$(or $\mathbb{C}$). Then up to some equivalence relations, the shape is determined by a finite number of *cone points* whose neighborhood is modeled by gluing a circular sector by its two edges.

We can further require the vertex of a triangulation to be placed at integral points of $\mathbb{R}^2$ and the cone points to be vertex. It can be formally defined as follows:

Definition 3.1. An *integral-affine structure* on a real surface S is a collection of charts from S to $\mathbb{R}^2$ such that the transition functions lie in $SL_2(\mathbb{Z}) \ltimes \mathbb{R}^2$. The *singularities* are those points where the integral-affine structure fails to extend.

Definition 3.2. An integral-affine surface is *triangulated* if it admits a triangulation into lattice triangles of lattice volume 1 (Euclidean volume 1/2) such that each singularity is a vertex of the triangulation.

Then a largest open chart with the lattice triangles can give rise to a triangulated integral-affine sphere. Locally the triangulation is determined by the following integers.

Definition 3.3. Let e_{ik} be a directed edge in the interior of a triangulated integral-affine surface. Let e_{ij} and e_{il} be the edges emanating from v_i directly clockwise and counterclockwise to e_{ik}. We define the *negative self-intersection* d_{ik} of the edge e_{ik} by the formula

$$d_{ik}e_{ik} = e_{ij} + e_{il},$$

where we view the edges as lattice vectors in some chart. Note that d_{ik} is an integer because (e_{ij}, e_{ik}) and (e_{ik}, e_{il}) are both oriented lattice bases and d_{ik} is independent of the choice of integral-affine chart.

Proposition 3.4. *[Eng18, Prop.3.6, 3.7] Let S be a triangulated integral-affine surface with singularities. The formula $d_{ik} + d_{ki} = 2$ holds for all interior edges e_{ik}. Any triangulated integral-affine surface is uniquely determined by the data of a collection of negative self-intersections d_{ik} for each directed interior edge e_{ik} such that $d_{ik} + d_{ki} = 2$.*

Example 3.5. Let the polytope in Figure 1 be an integral affine surface, then $d_{12} = -1$ and $d_{21} = 3$.

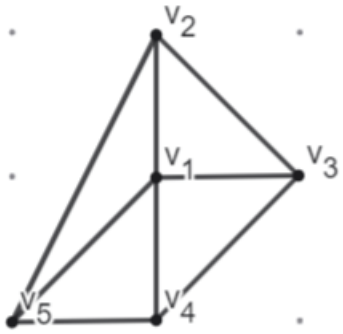


FIGURE 1. A pseudo-fan of the toric pair $(\mathrm{Bl}_1\mathbb{P}^2, D)$ with $(d_i) = (-1, 0, 1, 0)$.

As the dual complex of a Kulikov surface, each vertice should correspond to an anticanonical pair, each edge to a double rational curve and each triangle to a triple point. We start by the 'local' combinatorial model of a single anticanonical pair.

Definition 3.6 (Pseudo-fan)**.** Let $(V, D = D_1 + \cdots + D_n)$ be an anticanonical pair. The pseudo-fan $\mathfrak{F}(V, D)$ of (V, D) is a triangulated integral-affine surface whose underlying triangulated surface is the cone over the dual complex of D. By the lemma above, it suffices to declare that the negative-self intersection of the directed edge e_i pointing from the cone point to the vertex corresponding to a component D_i is

$$d_i = \begin{cases} -D_i^2, & n > 1 \\ 2 - D_i^2. & n = 1 \end{cases} \tag{3.1}$$

The imposed integral-affine structure has at most one singularity, at the cone point.

Here n=1 when D is a nodal rational curve. When $n > 1$, the equation 3.1 is the same as the relation between the rays in the toric fan of a toric surface. So the pseudo-fan of a toric pair can be constructed by directly taking the convex hull of the primitive vectors generationg the rays of its toric fan. We get a polygon by doing so.

However for a non-toric pair, the local chart of cone point of the pseudo-fan should look like a gluing of cirlular sector and the integral-affine structure cannot extend to the cone point. It will be studied in the next section.

Moreover, equations 2.2 and 3.1 implies that the pseudo-fans of those toric components of a Type III Kulikov surface can glue with each other in the way the pairs intersect.

3.2. **IAS^2 and Type III Kulikov Surfaces.** In order to study the integral-affine singularities corresponding to non-toric anticanonical pairs, we define a finer structure. Firstly we decribe the local model of a 'standard' singularity.

Definition 3.7. An I_1 singularity is the germ of a singular integral-affine surface isomorphic to the following basic example: Cut from $\mathbb{R}^2 = \mathbb{R}e_1 \oplus \mathbb{R}e_2$ the convex cone with the sides $\mathbb{R}_{\geq 0}e_2$

and $\mathbb{R}_{\geq 0}(e_2 - e_1)$ and glue one boundary ray to another by a transformation given by $e_1 \mapsto e_1$, $e_2 \mapsto -e_1 + e_2$.

In this definition, such a chart of the germ of a singularity is determined by 'cutting' a ray emanating from the singularity in $\mathbb{R}^2$ and separate the two sides by the monodromy (the way they glue). Canonically we can choose this ray to be the invariant one (up to sign) under the monodromy and clearly this *monodromy-invariant direction* is independent of the choice of 'cut'. Such a transformation along this canonical 'cut' is called a *shear*. This should be considered as a part of the definition of an I_1 singularity and also offers geometric information in the model.

Naturally we can have I_1 singularities collide then:

Definition 3.8. Let $\vec{v}_1, \cdots, \vec{v}_k$ be a sequence of primitive integral vectors, ordered cyclically counterclockwise around the origin. Define an integral-affine singularity $(S, p) = I(n_1\vec{v}_1, \cdots, n_k\vec{v}_k)$ to be the result of shearing the affine structure of $\mathbb{R}^2$ a total of n_i times along $\mathbb{R}_{\geq 0}\vec{v}_i$.

The singularity $I(n_1\vec{v}_1, \cdots, n_k\vec{v}_k)$ is the collision of $n_1 + \cdots + n_k$ I_1 singularities, with monodromy invariant directions along the $\vec{v}_i$. Instead of multiplicity or order, this number is called charge because of its geometric meaning.

Definition 3.9. The charge of a singularity (S, p) is the number $\sum_{i=1}^{k} n_i$ of rays sheared to produce it, counted with multiplicity. For instance, the I_1 singularity $I(\vec{v})$ has charge one.

Definition 3.10. An IAS^2 (to distinguish from integral-affine sphere), is a sphere $B = S^2$ and a finite set $\{p_1, \cdots, p_n\} \in B$ such that $B \setminus \{p_1, \cdots, p_n\}$ has a non-singular integral-affine structure, and a neighborhood of each p_i is modeled by some integral-affine singularity $I(n_1\vec{v}_1, \cdots, n_k\vec{v}_k)$.

Proposition 3.11. *The sum of charges of singularities of an IAS^2 is* 24.

Proof. See [Eng18]. □

Now a largest open chart of a triangulated IAS^2 contains more information than simply an triangulated integral-affine sphere in last section. That is, to each singularity we attach the shearing direction by a dashed ray. Recall our definition of a pseudo-fan of an anticanonical pair in last section, with the shear added to the triangles now we can determine the geometry of the non-toric pair. We firstly see the effect of the 'surgeries', say node smoothing and internal blow-up, on the pseudo-fan.

Proposition 3.12. *Let $(\widetilde{V}, \widetilde{D})$ be a deformation of an anticanonical pair (V, D) such that $\widetilde{D}$ is the smoothing of the node $D_{i-1} \cap D_i$. The pseudo-fan of $(\widetilde{V}, \widetilde{D})$ is the result of the following surgery on the pseudo-fan of (V, D):* ***Collapse the triangular face with edges e_i and e_{i+1} to a single edge with negative self-intersection*** *$d_i + d_{i+1} - 2$. The resulting shearing direction is parellel to the boundary component corresponding to the node $D_{i-1} \cap D_i$.*

Proposition 3.13. *Let $(\widetilde{V}, \widetilde{D})$ be an internal blow-up of an anticanonical pair (V, D) on the component D_i. The pseudo-fan of $(\widetilde{V}, \widetilde{D})$ is the result of the following surgery on the pseudo-fan of (V, D):* ***Keep the integral-affine structure fixed away from the edge e_i and increase the negative***

self-intersection of e_i ***from*** d_i ***to*** $d_i + 1$. *This surgery shears the pseudo-fan along* e_i *and the monodromy makes the two boundary components of* $D_{i-1} \cap D_i$ *and* $D_i \cap D_{i+1}$ *together a larger one.*

Both surgeries increase the charge by 1 as a unit shear. Moreover, a corner blow-up adds a edge and doesn't change the integral-affine structure or charge (but changes the triangulation!).

Remark 3.14. By [GHK15], any anticanonical pair, after some corner blow-ups, can be then blow down to a toric pair. Therefore one can construct its pseudo-fan by corner blow-ups based on a toric fan and any singularity of a triangulated IAS^2 is such a model.

Finally, we get a IAS^2 as the dual complex of a Type III Kulikov surfaces. Denote $\mathrm{Star}(v)$ the union of triangles containing the vertex v. We have the following correspondence.

Proposition 3.15. *[Eng18] The dual complex* $\Gamma(\mathcal{X}_0)$ *of a Type III degeneration of K3 surfaces admits a natural structure of an* IAS^2 *such that*

$$Star(v_i) = \mathfrak{F}(V_i, \sum_j D_{ij}).$$

Conversely, given a triangulated integral-affine sphere B*, there is a Type III degeneration* $\mathcal{X} \to \mathcal{C}$ *such that* $\Gamma(\mathcal{X}_0) = B$.

Proof. We can construct a 'Type III surface' with dual complex B by 'gluing' anticanonical pairs. The gluability follows from equations 2.2 and 3.1. □

3.3. **Modifications and base change.** As studied in last section, surgeries on anticanonical pairs will or will not affect the integral-affine singularity of a pseudo-fan. In this section we study modifications of Type III degenerations and their effects on the IAS^2 $\Gamma(\mathcal{X}_0)$.

Definition 3.16. [FM83b]

Let $\mathcal{X} \to (C, 0)$ be a Kulikov family and $E \subset \mathcal{X}_0$ be a smooth rational curve. Then there are birational modifications on $\mathcal{X}$ with center E, called elementary modifications.

(1) Soppose $E \cap (\mathcal{X}_0)_{Sing} = \varnothing$, $E^2 = -2$ on a component of $\mathcal{X}_0$, and that E does not extend to a Caritier divisor on $\mathcal{X}$. A *M0 Modification* is to contract E on the threefold $\mathcal{X}$ and resolve the resulting double point to obatin $\mathcal{X} \dashrightarrow \mathcal{X}'$. This does not change the isomorphism type of $\mathcal{X}_0$.

(2) Suppose E meets $(\mathcal{X}_0)_{Sing}$ in one point and $E^2 = -1$ on a component of $\mathcal{X}_0$. A *M1 Modification* is to blow up the threefold $\mathcal{X}$ along E and contract the induced exceptional divisor $\mathbb{P}^1 \times \mathbb{P}^1$ along another ruling, yielding $\mathcal{X}'$. On $\mathcal{X}_0$, the curve E is 'flipped' to E'.

(3)A *M2 modification* is to apply such flop to $E = D_{ij} = D_i \cap D_j$(in Type III case only). On $\mathcal{X}_0$, it is to blow down E on both V_i and V_j, blow up the two triple points T_{ijk} and T_{ijl} on the components V_k and V_l and the glue the resulting exceptional curves.

These birational modifications doesn't change the monodromy invariant because the punctured family $\mathcal{X}^* \to C^*$ is unchanged. Obviously M0 modifications do not change the IAS^2 $\Gamma(\mathcal{X}_0)$ and the effects of M0 and M1 Modifications are descirbed as following:

Definition 3.17. Let p be an integral affine singularity of an IAS^2 and $\vec{v}$ be a primitive vector along one of p's shearing directions. A nodal slide along $\vec{v}$ of length t is a surgery on the integral affine structure which translates by $t\vec{v}$ the originating point of one shearing ray in the direction $\vec{v}$.

The surgery does not change the integral-affine stucture on the complement of the segement $t\vec{v}$ and result in an I_1-singularity at the endpoint of $t\vec{v}$ (maybe colliding into another singularity).

Proposition 3.18. *An M1 modification preserves the triangulation of* $\Gamma(\mathcal{X}_0)$ *but changes the IAS*^{2} *by a unit length nodal slide, moving an* I_1*-singularity along* e_{ij} *from* v_i *to* v_j*.*

An M2 modification does not change the IAS^2 *structure on* $\Gamma(\mathcal{X}_0)$ *but retriangulates it by replacing the diagonal of an integral-affine unit square by the opposite one.*

For a Kulikov model $\mathcal{X} \to (C, 0)$, take the standard resolution of the order k base change $\mathcal{X}' \to (C', 0)$ ramified over 0, and then by [Fri83] we get a new Type III degeneration $\mathcal{X}[k]$ of which we can describe the new IAS^2 $\Gamma(\mathcal{X}_0[k])$.

Proposition 3.19. *The effect of order* k *base change on the dual complex* $\Gamma(\mathcal{X}_0)$ *is to take the standart refinement of every triangle into* k^2 *triangles each of which is standard lattice triangle of area* $\frac{1}{2}$*.*

3.4. **IAS^2 via symplectic geometry and the monodromy theorem.** To construct a Kulikov surface, it now suffices to give the triangulated IAS^2 from the monodromy of degeneration. Mirror symmetry provides a solution to the problem in a beautiful way.

In symplectic geometry, an natural integral-affine structure appears on the base of a Lagrangian fibration $\mu : (S, \omega) \to B$ of a symplectic 4-fold S with respect to the form ω. One can obtain a chart of the integral-affine structure by integrating the symplectic form of S over lifts of paths in $B \setminus Sing(B)$ where $Sing(B)$ denotes the singular locus of the fibration.

In particular, let $\overline{T}$ be a complex toric surface, $\overline{L} \in \mathrm{Pic}(\overline{T}) \otimes \mathbb{R}$ an ample class, and $\overline{\omega}$ a symplectic form with $[\overline{\omega}] = \overline{L}$. The moment map $\mu_{\overline{T}} : (\overline{T}, \overline{\omega}) \to \overline{B}$ is a Lagrangian torus fibration which induces the integral-affine structure on the moment polytope $\overline{B}$ coming from its embedding into $\mathbb{R}^2$. It degenerates over the toric boundary $\overline{D} \subset \overline{T}$, and sends the components of $\overline{D}$ to the boundary components of $\overline{B}$.

Following Symington [Sym02], when the generic fiber of μ is a smooth 2-torus but interior fibers may also degenerate to necklaces of spheres at some finite set of points, we can obtain a singular integral-affine structure on the disk B by *Symington surgeries*.

Definition 3.20. Let P_i be the components of the polygonal boundary of a chart of B.

An *internal blow-up* is the result of cutting a triangle of lattice size n (and lattice volume n^2) from the side of the moment polytope B corresponding to the component blown up, then gluing the two remaining edges, introducing an I_1 singularity $p \in B$ at the interior corner of the triangle.

A *node smoothing* of B at $v_{i,i+1} := P_i \cap P_{i+1}$ is the result of cutting along the segment from $v_{i,i+1}$ to a point $v := v_{i,i+1} + n(x_i + y_i)$ lying in $B - \partial B$ and then gluing the edges by the shearing map which fixes the line containing the cut and maps x_i to $-y_i$, where x_i, y_i denote the primitive integral vectors emanating from $v_{i,i+1}$ along P_{i+1} and P_i respectively. It introducing an I_1 singularity at v.

The positive integer n is called the size of the surgery. In particular, the chart defines B as an IAS^2 when (S, ω) is a K3 surface. Comparing Symington surgeries with Proposition 2.20 and 2.21, we see that an internal blow-up on the symplectic geometric side corresponds to a node smoothing on Star(p) and vice versa.

Remark 3.21. Let $\phi : T \to \overline{T}$ be a blow-up at a smooth point of the boundary $\overline{D}$ satisfying $[\omega] = \phi^*[\overline{\omega}] - nE$, with exceptional divisor E. Then the singular integral-affine disk B is the result of an internal blow-up of size n on the boundary of $\overline{B}$ corresponding to the component blown up. The correspondence is similar for the node smoothing. (Compare Symington surgeries with the surgeries on pseudo-fans in last section!)

Remark 3.22. In other mirror symmetry literatures, say [KS06] [GS06] [GS10], the authors study the collapsing of the Ricci-flat (hyperKahler) metric along the fibration. The integral-affine surface obtained is the Gromov-Hausdorff limit as a metric space with singularities then. One can regard this as a 'degeneration' or 'tropicalization' in the symplectic geometry setting.

Conversely, given an integral-affine manifold B, there is a Lagrangian torus fibration

$$T^*B/T^*_{\mathbb{Z}}B \to B,$$

where $T^*_{\mathbb{Z}}B$ is a lattice in the cotangent bundle T^*B. There is a nutural symplectic form defined from the local coordinates. When B is an integral-affine surface with singularities, by [Sym02] this can also be generalized, obtaining some singular fibers over integral-affine singularities of B.

In particular, if B is a generic IAS^2, there is a symplectic 4-manifold (S, ω) diffeomorphic to a K3 surface, together with $\mu : (S, \omega) \to B$ a Lagrangian torus fibration over B that has 24 singular fibers over the I_1 singularities.

From now on let B be a generic IAS^2. Let B^o be the nonsingular locus of B and $i : B^o \hookrightarrow B$ the inclusion. Then $T^*_{\mathbb{Z}}B := i_*(T^*_{\mathbb{Z}}B^o)$ is a constructible sheaf. Let $\gamma = \{(\gamma_i, \alpha_i)\}$ be a 1-cycle valued in the integral cotangent sheaf $T^*_{\mathbb{Z}}B$. Concretely, it is a collection of paths γ_i with constant covector fields α_i along γ_i such that at the boundaries of the paths, the vectors α_i add to zero in $T^*_{\mathbb{Z}}B$. When the paths γ_i are incident to an I_1 singularity, the covectors α_i must sum to a covector vanishing on the monodromy-invariant direction.

Definition 3.23. (Visible surfaces) From such a 1-cycle γ, we may construct a *visible surface* $\Sigma_\gamma \subset S$ inside the symplectic 4-manifold $\mu : (S, \omega) \to B$, by taking a cylinder in S which maps to γ_i whose fibers are the circles in the torus fiber that correspond to α_i via the symplectic form. Then the ends of these cylinders over the points in $\cup_i \partial\gamma_i$ are null-homologous in the fiber. Thus, we may glue in a (Lagrangian) 2-chain contained in the fiber over $\cup_i \partial\gamma_i$ and produce a closed PL surface Σ_γ.

Example 3.24. The simplest example of a visible surface is a path connecting two I_1 singularities with parallel monodromy-invariant lines (under parallel transport along the path).

Note that $[\Sigma_\gamma] \cdot f = 0$, where $f = [\mu^{-1}(\mathrm{pt})]$ is the Lagrangian fiber class. We do have a well-defined class $[\Sigma_\gamma] \in f^\perp/f$.

There is an important special case of the above construction.

Example 3.25. Suppose that all γ_i's are straight line segments e_{ij} forming a graph in the integral-affine structure on B, and that the tangent vector field is an integer multiple n_{ij} of the primitive integral covector vanishing along γ_i. Then the cylinder lying over e_{ij} can be made Lagrangian and the surface Σ_γ is a PL Lagrangian surface in (S, ω). We call the result a *Lagrangian visible surface*.

The integral affine geometry in sympletic geometry finally helps understand the monodromy invariant $\lambda \in \delta^\perp/\delta$.

Theorem 3.26. *Let B be a generic triangulated IAS^2, then*

(1) Let $\mathcal{X} \to C$ be a type III Kulikov degeneration such that $\Gamma(\mathcal{X}_0) = B$.

(2) Let $\mu\colon (S, \omega) \to B$ be a Lagrangian torus fibration over the same B.

Then there exists a diffeomorphism $\phi\colon S \to \mathcal{X}_t$ to a nearby fiber $t \neq 0$ such that

(a) $\phi_* f = \delta$,

(b) $\phi_*[\omega] = \lambda$ in $\delta^\perp/\delta \otimes \mathbb{R}$.

Proof. The diffeomorphism is similar to the one constructed in [EF21] for degenertation of Type III anticanonical pairs: the Clemens collapse gives the nearby fiber $\mathcal{X}_t$ the same structure of torus fibration over the intersection complex $\Gamma(\mathcal{X}_0)^\vee$ (the dual of the dual complex $\Gamma(\mathcal{X}_0)$). Then (a) immediately follows.

For (b), we consider the collection of Lagrangian visible surfaces Σ_γ which fiber over the 1-skeleton $\Gamma(\mathcal{X}_0)^{[1]}$. It suffices to show the images under ϕ_* of the classes $[\Sigma_\gamma]$ generate a 19-dimensional lattice in $H^2(\mathcal{X}_t; \mathbb{Z})$ which is invariant under the Picard-Lefschetz transformation. Since $[\omega] \cdot [\Sigma_\gamma] = 0$, we conclude that the monodromy invariant λ and $\phi_*[\omega]$ are proportional in $\delta^\perp/\delta$. By [FS86],

$$\lambda^2 = \#\{\text{triple points of} \mathcal{X}_0\} = \text{vol}(\Gamma(\mathcal{X}_0)) = [\omega]^2.$$

□

Thus, the method of giving Kulikov surfaces from monodromy invariants is now clear. On the symplectic geometry side, it is more straightforward to construct the K3 surface with the corresponding intersection relationship of ω and Σ_γ.

3.5. **Example: Degeneration of elliptic K3 surfaces.** In the presence of a polarization $M \hookrightarrow \text{Pic}(\mathcal{X}^*)$ for K3 surfaces, where M is a primitive hyperbolic sublattice of signature $(1, r-1)$ with $r \leq 20$, we see that $T \in O(H^2(\mathcal{X}_t))$ lies in the subgroup Γ fixing M. In particular, the lattice of the monodromy invariants λ lies in a subquotient of $M^\perp$.

Let vectors s, f in Λ_{K3} satisfy $s^2 = -2$, $f^2 = 0$, and $(s, f) = 1$ and then $M :=< s, f >$ is the standard rank 2 even unimodulair lattice $U = \text{II}_{1,1}$ and $M^\perp = \text{II}_{2,18} \subset \Lambda_{\text{K3}}$. By choosing appropriate marking of $H^2(\mathcal{X}_t)$, $\mathcal{X}_t$ are elliptic K3 surfaces $\mathcal{X}_t \to \mathbb{P}^1$ with a section s and fiber f. Since there is one unique $O(\text{II}_{2,18})$-orbit of isotropic vector δ in $\text{II}_{2,18}$, the monodromy invariant λ lies in $\delta^\perp/\delta \simeq \text{II}_{1,17}$.

Consider the positive cone C in $\text{II}_{1,17}$ which is invariant under the action of the isometry group $\Gamma = O^+(\text{II}_{1,17})$ of spinor norm 1. Then Γ contains an index 2 subgroup W generated by reflections in the roots, the (-2)-vectors $\alpha \in \text{II}_{1,17}$. There is a choice of 19 simple roots generationg W and the fundamental chamber $\mathfrak{K} = \{\lambda \in \overline{C}_{\mathbb{Q}} | \lambda.\alpha_i \geq 0\}$ has 19 facets. The Coxeter diagram of the 19 roots is given in Fig.2.

In [ABE22] the author constructed a Type III Kulikov model for any monodromy invariant $\lambda \in \mathfrak{K}$ via integral-affine geometry as introduced in previous sections. We reviewed the result here without explanation.

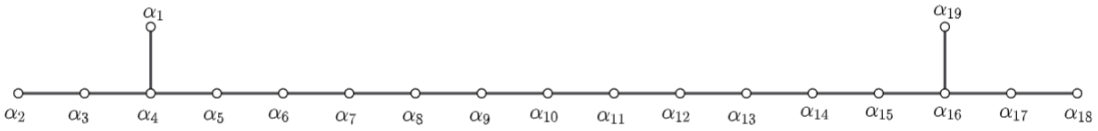


FIGURE 2. Coxeter diagram of $\mathrm{II}_{1,17}$

A subdivision of $\mathfrak{K}$ is constructed by adding the hyperplanes perpendicular to the vectors

$$\beta_L = \alpha_3 + 2\alpha_2 - \alpha_1, \gamma_L = \alpha_3 - \alpha_1, \beta_R = \alpha_{17} + 2\alpha_{18} - \alpha_{19}, \gamma_R = \alpha_{17} - \alpha_{19}.$$

Here L and R represent left and right sides of the diagram, and the hyperplanes of the two sides work the same way and independently of each other. Each side divides $\mathfrak{K}$ into three maximal cones, defined by $\{-\lambda.\beta \geq 0\}$, $\{\lambda.\beta \geq 0, -\lambda.\gamma \geq 0\}$ and $\{\lambda.\gamma \geq 0\}$. We say L(or R)$= 1, 2$ and 3 respectively.

Write $a_i = \alpha_i \cdot \lambda$, $b_L = \beta_L \cdot \lambda$, $c_L = \gamma \cdot \lambda$, $b_R = \beta_R \cdot \lambda$, $c_R = \gamma_R \cdot \lambda$. Then any $\lambda \in \mathfrak{K}$ is characeterized 19 non-negative numbers $(\ell_i) \in \mathbb{Q}^{19}$, given by the following table:

L End	ℓ_1	ℓ_2	ℓ_3	$\cdots$	ℓ_i	$\cdots$	ℓ_{17}	ℓ_{18}	ℓ_{19}	R End
1	$-b_L/3$	a_2	a_3				a_{17}	a_{18}	$-b_R/3$	1
2	$b_L/2$	$-c_L/2$	a_3	$\cdots$	a_i	$\cdots$	a_{17}	$-c_R/2$	$b_R/2$	2
3	a_2	$c_L/2$	a_1				a_{19}	$c_R/2$	a_{18}	3

Define 19 vectors in $(\frac{1}{2}\mathbb{Z})^2$ by

$$\vec{v}_1 = \begin{cases} (0,1) & \text{if } L = 2,3 \\ (1, \frac{9}{2}), & \text{if } L = 1 \end{cases}$$

$$\vec{v}_i = (1, \tfrac{10-i}{2}) \text{ if } i = 2, \ldots, 18$$

$$\vec{v}_{19} = \begin{cases} (0,-1) & \text{if } R = 2,3 \\ (1, -\frac{9}{2}) & \text{if } R = 1. \end{cases}$$

Proposition 3.27. *For any $\lambda \in \mathfrak{K}$, $\sum \ell_i(\lambda)\vec{v}_i$ is a horizontal vector.*

Form a polygon $P(\lambda)$ whose edges are the vectors $\ell_i(\lambda)\vec{v}_i$ put end-to-end in the plane, together with a segment on the x-axis. See Fig.3. Let $Q(\lambda)$ be the lattice polygon which results from taking the union of $P(\lambda)$ with its reflection across the x-axis.

Definition 3.28. Define $B(\lambda)$, an integral affine sphere, as follows: Glue each edge $\ell_i\vec{v}_i$ of $Q_{LR}(\ell)$ to its reflected edge by an element of $\mathrm{SL}_2(\mathbb{Z}) \ltimes \mathbb{R}^2$ which preserves vertical lines. This uniquely specifies the gluings, except when $\ell_1, \ell_{19} > 0$ and $L, R \in \{2,3\}$ respectively. For these edges, we must specify the gluing to be $-A^4$ where $A(x,y) = (x+y, y)$ is a unit vertical shear.

The vertices of $Q(\lambda)$, after glued, become integral affine singularities of $B(\lambda)$. The monodromy of every one of them is in $\mathrm{SL}_2(\mathbb{Z})$. The integral affine structures of $B(\lambda)$ are the same when we interchange the end behavior 2 and 3.

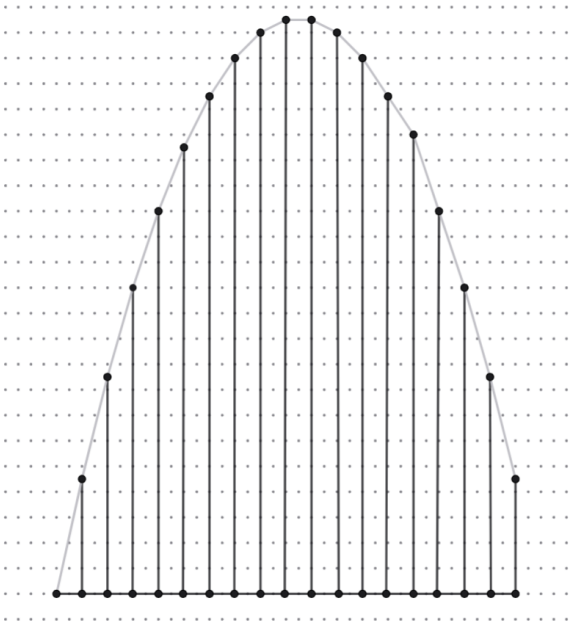


FIGURE 3. Polygon $P(\lambda)$ of $(\mathrm{IAS}^2, R_{\mathrm{IA}})$ for a Type III Kulikov surface. End behaviors: $L = 1, R = 2$ or 3.

Now we define $B(\lambda)$ an IAS^2, giving the monodromy invariant directions (recall Def 3.8) of each singularity by rays emanating from it(TABLE 2. *loc.cit.*):

L	Singularity	Marked rays	Notation
1	$\ell_1 \neq 0$, end singularity	$(1,-3),(1,0),(1,3)$	X_3
$1,2$	$\ell_1 = 0$, $\ell_2 \neq 0$	$(1,-2),(1,0),(1,1),(1,3)$	X_4
3	$\ell_1 = 0$, $\ell_2 \neq 0$	$(1,-2),(1,0),(1,2),(1,4)$	X_4'
$1,2,3$	$\ell_i = 0$ for $i \leq k$, $k \geq 2$	All choices equivalent	X_{k+3}
$2,3$	$\ell_1 \neq 0$, inner singularity	$(1,0),(1,2)$	Y_2
2	$\ell_1, \ell_2 \neq 0$, outer singularity	$(1,1),(1,3)$	Y_2
3	$\ell_1, \ell_2 \neq 0$, outer singularity	$(1,2),(1,4)$	Y_2'
$2,3$	$\ell_1 \neq 0$, $\ell_i = 0$ for $2 \leq i \leq k$	All choices equivalent	Y_{k+1}
	$\ell_{i+j} = 0$, $1 \leq j \leq k$ in interior	$(0,-1)$, multiplicity k	I_k

When an end is an isolated point, the symbol X is used. When the end is a vertical segment the symbols Y are used for the so-called *inner* and *outer* singularities at the points p_1 and p_2, respectively. The same applies to p_{20} and p_{19} at the right end.

One can describe the *combinatorial type* of $B(\lambda)$ by listing all singularities on it with notations X, I, Y's. For instance, if $(L, R) = (3, 2)$ and $\ell_i \neq 0$ for exactly $i = 2, 5, 6, 16, 19$ then we say that $B_{LR}(\ell)$ is of combinatorial type $X_4' I_3 I_1 I_{10} Y_4 Y_2$ indicating the sequence of singularities one sees traveling along the vectors $\ell_i \vec{v}_i$.

Definition 3.29. Decomposing the chart $Q(\lambda)$ into unit width vertical strips and triangulating each trapezoid completely into unit lattice triangles makes $B(\lambda)$ a triangulated IAS^2. Define $\mathcal{X}_0(\lambda)$ to

be the unique deformation type of Type III Kulikov surface associated to the triangulated $B(\lambda)$ by Theorem 3.15.

Together with the Kulikov surface, this model includes a polarizing divisor $R = s + m\Sigma f_i$, where s is the section and f_i are the fibers of the elliptic fibration, counted with multiplicities on every $\mathcal{X}_0(\lambda)$. Recall that a line bundle $\mathcal{L} \to \mathcal{X}$ on a Kulikov model satisfies a compatibility condition $L_i \cdot D_{ij} = L_j \cdot D_{ji}$, where $L_i := \mathcal{L}\big|_{V_i} \in \mathrm{Pic}(V_i)$.

Definition 3.30. Let B be an IAS^2. An *integral-affine divisor* R_{IA} on B consists of two pieces of data:

(1) A weighted graph $R_{\mathrm{IA}} \subset B$ with vertices v_i, straight line segments as edges e_{ij}, and integer labels n_{ij} on each edge.
(2) Let $v_i \in R_{\mathrm{IA}}$ be a vertex and (V_i, D_i) be an anticanonical pair such that $\mathfrak{F}(V_i, D_i)$ models v_i and contains all edges of e_{ij} coming into v_i. We require the data of a line bundle $L_i \in \mathrm{Pic}(V_i)$ such that $\deg L_i \cdot D_{ij} = n_{ij}$ for the components D_{ij} of D_i corresponding to edges e_{ij} and L_i has degree zero on all other components of D_i.

The underlying weighted graph R_{IA} of R in $B(\lambda)$ is the union of the following straight lines:

(1) the horizontal line joining the two ends, with label $n_{ij} = 1$, and
(2) the vertical line through any singularity, with label $n_{ij} = Q$, where Q is the total charge of the singularities on the vertical line.

Then, on each component of $\mathcal{X}_0(\lambda)$ corresponding to a vertex of R_{IA}, the explicit description of the chosen line bundle is given in [ABE22, Sec. 7B].

The construction of Kulikov surfaces $\mathcal{X}_0$ and polarizing divisors R give the construction of Kulikov models of Type III degeneration of polarized elliptic K3 surfaces. We conclude the main results as follows.

Theorem 3.31. *[ABE22, Prop 7.23] Let $\mathcal{X}(\lambda) \to C$ be a smoothing of an elliptically fibered $\mathcal{X}_0(\lambda)$ which keeps f and s Cartier. Then the general fiber is an elliptic K3 surface, the fibers in the polarizing divisor R are the limits of the singular fibers, and the section s is the limit of the section.*

Theorem 3.32. *[ABE22, Cor. 7.33] The monodromy invariant of $\mathcal{X}(\lambda)$ is the unique lattice point $\lambda \in \mathfrak{K}/\mathbb{Z}_2$, where $\mathbb{Z}_2$ is generated by the action of exchanging the indices L and R.*

Remark 3.33. We omitted many details in the construction, such as the choice of the surface $\mathcal{X}_0$ in the deformation type given by the IAS^2 $B(\lambda)$ and the reason of choosing the polarizing divisor $R = s + m\Sigma f_i$.

Remark 3.34. Such procedure is kind of a 'standard' way to construct Kulikov models of polarized K3 surfaces with pre-given monodromy invariant, in order to study the toroidal compactification of moduli of K3 surfaces polarized by any lattic M, using the theory of IAS^2. See also [AE23] for another example.

4. Type II degeneration

4.1. **Converting to Type III degeneration of ellipitic K3 surfaces.** In past works, Type II models are regarded as limiting cases of Type III under the framework of IAS^2, and are constructed case-by-case for low-degree K3 surfaces. Now we introduce another strategy to study the Type II degeneration of K3 surfaces. The idea is to convert the problem on Type II degeneration into Type III degeneration of elliptic K3 surfaces. We have the following theorem for embedding of lattices.

Theorem 4.1. *[Nik79, Thm 1.14.4]*

Let M be an even lattice of signature (t_+, t_-) and let L be an even unimodular lattice of signature (l_+, l_-). There exists a unique primitive embedding of M into L if $t_\pm < l_\pm$ and

$$l(M) + 2 \leq rk(L) - rk(M),$$

where $l(M)$ denotes the the minimal number of the generators of the discriminant group G_M.

Consider a Type II degeneration of primitively polarized K3 surfaces as determined by an rank 2 isotropic sublattice I of Λ_{K3} together with the primitive polarization class $v \in \Lambda_{\mathrm{K3}}$ such that $(v, v) = 2d$ and $(I, v) = 0$.

Definition 4.2. We call such a pair (I, v) a *Type II pair*.

Fixing such an I, by the unimodularity of Λ_{K3} one can embed I in $U \oplus U$ and get an splitting $\Lambda_{\mathrm{K3}} \simeq U \oplus U \oplus (I^\perp/I)$ of Λ_{K3} [Huy16, Cor 14.1.14]. Then $I^\perp/I$ must also be unimodular, hence isomorphic to the unique even unimodular lattice $\mathrm{II}_{1,17}$ of signature $(1, 17)$. By this identifictaion, $\overline{v} = v + I \in I^\perp/I$ determines a vector in $\mathrm{II}_{1,17}$ with $(\overline{v}, \overline{v}) = 2d$.

The automorphisms of Λ_{K3} that stablize I must preserve both I and $I^\perp$ and thus induce automorphisms on the quotient lattice $I^\perp/I \simeq \mathrm{II}_{1,17}$. On the other hand, since any automorphism of $\mathrm{II}_{1,17}$ can be extended to an automorphism of Λ_{K3} by acting trivially on I, all automorphisms in $O(\mathrm{II}_{1,17})$ can be induced in this way. In conclusion, we obtain a surjective map ϕ from the $O(\Lambda_{\mathrm{K3}})$-orbits of Type II pairs to the $O(\mathrm{II}_{1,17})$-orbits of vectors $\overline{v}$ of self-product $2d$.

Remark 4.3. Since the combinatorial type of a Type III degeneration of elliptic K3 surfaces is in particular given by the class of the monodromy invariant $\lambda \in \delta^\perp/\delta = \mathrm{II}_{1,17} \bmod \Gamma = O^+(\mathrm{II}_{1,17})$. Particularly (λ, λ) equals the number of triple poionts in the central fiber. By this one can regard the map ϕ as between classes of two different types of degenerations of K3 surfaces.

Conversely, suppose we have a vector $\overline{v} \in \mathrm{II}_{1,17}$ with $(\overline{v}, \overline{v}) = 2d > 0$. By Nikulin's theorem above, there exists a primitive embedding $\psi : \mathrm{II}_{1,17} \hookrightarrow \Lambda_{K3}$ unique up to automorphisms of Λ_{K3}. Again by unimodularity we have isomorphism $\psi(\mathrm{II}_{1,17})^\perp \simeq U \oplus U$. Choose any primitive rank-2 isotropic sublattice $I \subset U \oplus U$ and a lift $v = \psi(\overline{v}) + e, e \in I$ of $\overline{v}$, we get a Type II pair $(I, \psi(\overline{v}) + e)$. Since there is only one $O(U \oplus U)$-orbit of such isotropic sublattice I in $U \oplus U$, plus the uniqueness of ψ we only need to see when two pairs $(I, \psi(\overline{v}) + e_1)$ and $(I, \psi(\overline{v}) + e_2)$ are in the same $\mathrm{Stab}_{O(\Lambda_{K3})}(I)$-orbit of such pairs.

Definition 4.4. For an even lattice L and any $e, a \in L$, such that e is primitive isotropic and $a \in e^{\perp}$. The *elementary (Eichler) transformation* $t(e,a)$ is defined by

$$t(e,a) : v \mapsto v - (v,a)e + (v,e)a - \frac{1}{2}(a,a)(v,e)e$$

and is an isometry of L.

Lemma 4.5. *Let $\{x_1, \cdots, x_{22}\}$ be a $\mathbb{Z}$-basis of Λ_{K3} such that $\{x_1, x_2\}$, $\{x_3, x_4\}$ are canonical basis of two hyperbolic planes and hence $\{x_1\}_{i=5}^{22}$ is a basis for $L = II_{1,17}$. Let $I =< x_1, x_3 >$, then*

$$Stab_{O^+(\Lambda_{K3})}(I) =< t(x_1,x_4), t(x_2,x_3), t(x_1,x_3), t(x_1,v), t(x_3,v), O^+(L) | v \in L >$$

Proof. See [GHS09]. □

Lemma 4.6. *When $\overline{v}$ is primitive in $II_{1,17}$, all lifts are in the same $Stab_{O(\Lambda_{K3})}(I)$-orbit.*

Proof. We keep using notations in Lemma 4.5. For any primitive vector $\overline{v} \in \mathrm{II}_{1,17}$ lifted to $\overline{v} + e \in \mathrm{II}_{1,17} + I$ with $e \in I$ (we omit the ψ in the lift), take $v' \in \mathrm{II}_{1,17}$ with $(\overline{v}, v') = 1$ and then we can verify that $t(x_1, v')$ sends $\overline{v} + e$ to $\overline{v} + e - x_1$. By the same construction for $t(x_3, v')$ we see that any vectors $\overline{v} + e_1$ and $\overline{v} + e_2$ are in the same orbit. □

Lemma 4.7. *When $\overline{v} = k\overline{v}_0$ and $\overline{v}_0$ is primitive, for any $d|k$, there is exactly one $Stab_{O(\Lambda_{K3})}(I)$-orbit of lifts.*

Proof. Notice that any lift $k\overline{v}_0 + e$ of $k\overline{v}_0$ can only have imprimitivity dividing k in Λ_{K3}. Let $e = mx_1 + nx_3$, $m, n \in \mathbb{Z}$ and the primitivity is $\gcd(m,n,k)$ (take $\gcd(n,0) = n$).

Consider the transformation $t(x_1, v')$ where $v' \in \mathrm{II}_{1,17}$, it sends $k\overline{v}_0 + e$ to $k\overline{v}_0 + e - k(\overline{v}_0, v')x_1$. So we may assume $(\overline{v}_0, v') = 1$ and the result of $t(x_3, v')$ is similar. On the other hand, the transformation $t(x_1, x_4)$ fixes $\overline{v}, x_1$ and sends x_3 to $x_3 - x_1$, and similarly for $t(x_2, x_3)$. Thus any lift of imprimitivity $d|k$ can be transformed to $k\overline{v}_0 + dx_1$. So for any $d|k$, there is exactly one $\mathrm{Stab}_{O(\Lambda_{\mathrm{K3}})}(I)$-orbit of lifts. □

In particular, there is exactly one $\mathrm{Stab}_{O(\Lambda_{\mathrm{K3}})}(I)$-orbit of primitive lift of $\overline{v}$ when $\overline{v}$ is imprimitive. We conclude that:

Theorem 4.8. *There is a bijection between $O(\Lambda_{K3})$-orbits of Type II pairs (I,h) and $O(II_{1,17})$-orbits of vectors $\lambda \in II_{1,17}$ of positive norm.*

Corollary 4.9. *There is a bijection between the monodromy types of Type II degeneration of primitively polarized K3 surfaces of degree $2d$ and monodromy types of Type III degeneration of elliptic K3 surfaces whose Kulikov surface has $2d$ triple points.*

Remark 4.10. This idea of studying Type II pairs can be seen in another way. Extending the framework of algebraic polarized K3 surfaces, the moduli space $\mathcal{M}_{\mathrm{K3}}$ of all Kahler K3 surfaces parametrizes orbits of 3-dimensional positive definite subspace of $\Lambda_{\mathrm{K3}} \otimes \mathbb{R}$. The boundary of Satake compactification $\overline{\mathcal{M}_{\mathrm{K3}}}$ is stratified by components correspponding to (flags of) 1, 2, or 3-dimensional isotropic subspaces of $\Lambda_{\mathrm{K3}} \otimes \mathbb{Q}$. What we have done above is adding the polarization $c_1(L)$ as additional period and thus map the Type II cusps of $\overline{\mathcal{F}_{2d}}$ to some strata of $\overline{\mathcal{M}_{\mathrm{K3}}} \setminus \mathcal{M}_{\mathrm{K3}}$. [OO18, Section 6]

4.2. **Type II Kulikov surface via lattice polygon.** As reviewed in section 2.6, for any monodromy invariant $\lambda \in \mathrm{II}_{1,17}$ there is an IAS^2 $B(\lambda)$ giving the Type III Kulikov surface $\mathcal{X}_0(\lambda)$ for elliptic K3 surfaces. In this section, we construct a Type II Kulikov surface X_0 out of the underlying lattice polygon $Q(\overline{h})$ of $B(\overline{h})$ for $\overline{h} \in \mathrm{II}_{1,17}$.

4.2.1. *Construction of polarized Type II anticanonical pairs.* Such a lattice polygon $Q(\overline{h})$ is a reflection across the x-axis of the interior of a PL function. Each bending point of the function is a integral affine singularity of $B(\overline{h})$. By cutting vertically at these points, we decompose the polygon into strips, which are trapezoids or triangles on the two ends, and trapezoids in the middle. We call them *end strips* or *intermediate strips* respectively.

For every type of strip that might occur, we construct a Type II anticanonical pair (V, D) and a class of polarization $h \in H^2(V)$. Recall that a Type II anticanonical pair is a rational surface with an anticanonical elliptic curve E (components at the two end of a Kulikov surface), or a surface birational to $E \times \mathbb{P}^1$ with an anticanonical divisor consisting of two disjoint elliptic curves (components in the intermediate).

In our construction, the rational anticanonical pairs are either $\mathbb{P}^2$ or $\mathbb{P}^1 \times \mathbb{P}^1$ blown up on an anticanonical elliptic curve E, whose anticanonical divisor is the strict transform of E. On the other hand, starting from a $\mathbb{P}^1$-bundle over E with zero and infinity sections s_0, s_∞, the intermediate anticanonical pairs are obtained by blow-ups on the section s_∞, with the anticanonical divisor $D = D_0 + D_\infty$ being the strict transform of the elliptics s_0 and s_∞.

Notation 4.11. For simplicity, we place an end strip in a coordinate system, making it symmetric about the x-axis, to describe it by points A, B, E, or F. See Figure. 4. All allowable types of end strips and the polarized components $(V, D; h)$ are listed in Table. 1, correspondingly with the symbols of integral affine singularities described in 3.5. When the strip is a triangle, the smallest case is given, and only the polarization will be changed by a multiplication by k when the triangle is enlarged by k times. We describe the left end cases there and for the right end cases the construction is the same.

For intermediate strips, let the length of its left edge be d and the width of the trapezoid be m, $d, m \in \mathbb{Z}$. Suppose the slope of the upper edge of the strip left to it is k and the vertex E is an I_n-singularity. The surface will be determined by two integers, the degree s_0^2 and the number of blow-ups on s_∞, which are listed in the second column of Table. 2.

From now on, on the constructed surfaces, l represents the class of lines on $\mathbb{P}^2$, s, f represents classes of two rulings on $\mathbb{P}^1 \times \mathbb{P}^1$, and f, we abusing the notation, can also represent the class of ruling on the intermediate components. e_i represent the exceptional curves.

Remark 4.12. In the case of end strip Y_2Y_2', the polarization is well-defined because we have $\frac{m+d}{2} \in \mathbb{Z}$ by the construction of the polygon $Q(\lambda)$.

It's easy to check that h is numerically effective on V. Moreover, the combinatorial structure of such a lattice strip encodes geometric information of the triple $(V, D; h)$ constructed.

TABLE 1. Polarized Type II anticanonical pair $(V, D; h)$ for left end strips

Singularity	Shape of strip	Surface V	Class of D	Polarization h
$X_{n+3\geq 4}$	$A(0,0), \mathrm{B}(1, \frac{9-n}{2})$	$\mathrm{Bl}_n(\mathbb{P}^2)$	$3l - e_1 - \cdots - e_n$	$3l - e_1 - \cdots - e_n$
X_3	$A(0,0), \mathrm{B}(\frac{1}{3}, \frac{3}{2})$	$\mathbb{P}^2$	$3l$	l
X_4'	$A(0,0), \mathrm{B}(\frac{1}{2}, 2)$	$\mathbb{P}^1 \times \mathbb{P}^1$	$2s + 2f$	$s + f$
$Y_2Y_{n+2\geq 3}$	$\mathrm{E}(0,d), \mathrm{F}(m, d + m(4 - \frac{n}{2}))$	$\mathrm{Bl}_n(\mathbb{P}^1 \times \mathbb{P}^1)$	$2s + 2f - e_1 - \cdots - e_n$	$m(2s + 2f - e_1 - \cdots - e_n) + df$
Y_2Y_2	$\mathrm{E}(0,d), \mathrm{F}(\frac{m}{2}, d + 2m))$	$\mathbb{P}^1 \times \mathbb{P}^1$	$2s + 2f$	$m(s + f) + df$
Y_2Y_2'	$\mathrm{E}(0, \frac{d}{2}), \mathrm{F}(\frac{m}{2}, \frac{d}{2} + 2m))$	$\mathrm{Bl}_1(\mathbb{P}^1 \times \mathbb{P}^1)$	$2s + 2f - e_1$	$ms + \frac{3m+d}{2} f - me_1$

TABLE 2. Polarized Type II anticanonical pair $(V, D; h)$ for intermediate strips

	Surface	Classes of $D = D_0 + D_\infty$	Polarization h
the strip left to it is not Y_2Y_2'	$(-2k, n)$	s_0 and $s_\infty - e_1 - \cdots - e_n$	$df + ms_\infty - m(e_1 + \cdots + e_n)$
the strip left to it is Y_2Y_2'	$(-7, n-1)$	s_0 and $s_\infty - e_1 - \cdots - e_{n-1}$	$df + ms_\infty - m(e_1 + \cdots + e_{n-1})$

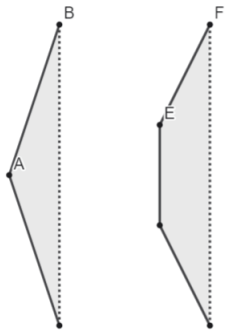


FIGURE 4. X, Y type left end strips cut from the lattice polygon

Proposition 4.13. *The integral-affine area (i.e. twice the Euclidean area) of a strip equals h^2. For left end strips, the length of the right edge equals $h \cdot D$. For intermediate strips, the length of left edge equals $h \cdot D_0$ and the length of right edge equals $h \cdot D_\infty$.*

Proof. Check by direct computation. □

4.2.2. *Construction of d-semistable Kulikov surfaces.* Now for any lattice polygon $Q(\overline{h})$ decomposed into r strips, fixing an elliptic curve E we get r polarized Type II anticanonical pairs $(V_i, D_i; h_i), r = 1, \cdots r$ ordered from left to right, such that all of the elliptic curves in D_i's are isomorphic to E. So the Kulikov pairs (V_i, D_i)'s can be glued, by translation of E, to a (not necessarily d-semistable) Type II Kulikov surface $\overline{X_0}(\overline{h}) = V_1 \cup \cdots \cup V_r$ along these elliptic curves. There is one special case requiring argument:

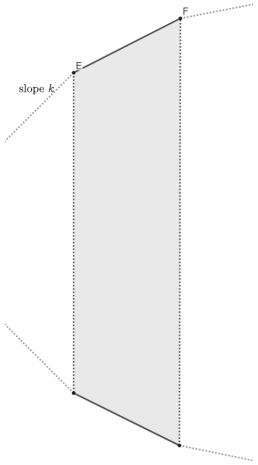


FIGURE 5. Intermidiate strips cut from the lattice polygon

When there is only one strip on $Q(\overline{h})$, say the left end, then the integral-affine area of this strip must be even by the evenness of $\mathrm{II}_{1,17}$. And one can check for each case that, this in turn restricts that the length of the right edge must be even, say $2d$. Then we define a right end polarized anticanonical pair $(V, D; h) = (\mathbb{P}^1 \times \mathbb{P}^1, E \in |2s + 2f|; df)$. This gives $h^2 = 0$ and $h.D = 2d$ and we can glue the left one to it.

Remark 4.14. Proposition 4.13 also holds for this case.

Definition 4.15. We say the left/right end component of a Type II Kulikov surface is of *E, or D-tyoe* if it is (blow-up of) $\mathbb{P}^2$ or $\mathbb{P}^1 \times \mathbb{P}^1$, respectively.

Lemma 4.16. *The double locus where the first $r-1$ components glue satisfy*

$$D^2_{i,i+1} + D^2_{i+1,i} = 0, i = 1, \cdots r-2.$$

Proof. We only need to consider the gluing of intermediate components to the component on its left. In our construction, the self-intersection of the anticanonical divisor in the left end components is $2k$ except for the case $Y_2Y'_{n+2}$, which is $2k-1$, where k is the slope of the upper edge of the strip. The same result applies to the self-intersection D^2_∞ in the intermediate components. Then the condition for self-intersection s_0^2 in the intermediate components implies $D^2_{i,i+1}+D^2_{i+1,i} = 0, i = 1, \cdots r-2$. □

To make $\overline{X_0}(\overline{h})$ d-semistable, we need to modify the self intersection of $D_{r-1,r}$ and $D_{r,r-1}$. Since $D^2_{r-1,r}$ equals twice the slope of the upper edge of the $(r-1)$-th strip and $D^2_{r,r-1}$ is the inverse of twice the slope of the right end slope, we can just blow up n points on D_r in V_r when the edge corresponding to the double locus is an I_n singularity, so that $D^2_{r-1,r} + D^2_{r,r-1} = 0$. Denote the resulting surface by $X_0(\overline{h})$. (Here we omit some discussion on the case when the $(r-1)$-th or r-th strip is $Y_2Y'_{n+2}$.)

Definition 4.17. We call these blow-ups *additional blow-ups*.

Remark 4.18. A d-semistable Kulikov surface has a total of 24 charges. Consider an E-type end of charge 3, a D-type end of charge 4 and each blow-up of charge 1, then the additional blow-ups can be understood as making the total charge 24.

Theorem 4.19. *There is a choice of construction such that the Kulikov surface $X_0(\overline{h})$ is d-semistable.*

Proof. Recall the description of deformation of Kulikov surface given in [AE23]. The topologically trivial deformation of $X_0(\overline{h})$ results from deforming the gluings of double curves and topologically trivial deformation of the anticanonical pairs.

In our construction, the choices of points blown up on every components and the translations of every gluing are completely free. So we can choose any Kulikov surface in this deformation space of $X_0(\overline{h})$. This space has large dimension $\mathrm{rk}\widetilde{\Lambda}(X_0(\overline{h}))$, and has a sublocus of dimension $\mathrm{rk}\Lambda(X_0(\overline{h}))$ consisting of d-semistable, topologically trivial deformation of $X_0(\overline{h})$. These are the Kulikov surfaces whose period points $\widetilde{\psi}_{X_0(\overline{h})}$ descend to $\mathrm{Hom}(\Lambda, E)$ as in prop 2.16. Therefore we can always get a d-semistable Kulikov surface $X_0(\overline{h})$. □

Notation 4.20. For clarity, corresponding to $(V_i, D_i; h_i)$, the subscript i of the notations l, s, f of class of curves now means the class is on the i-th component from left to right.

4.2.3. *Construction of polarized Kulikov models.* By the d-semistability of $X_0(\overline{h})$, there is a smoothing $X(\overline{h}) \to C$ of $X_0(\overline{h})$ into smooth K3 surfaces. It is a Type II Kulikov model. We want a polarization $L \to X(\overline{h})$ on this Kulikov model. Keeping $h_r \in H^2(V_r)$ unchanged by the additional blow-ups, we have

Proposition 4.21. *The collection of $h_i \in H^2(V_i)$ gives a Numerically Cartier class $(h_i) \in \Lambda(X_0(\overline{h}))$.*

Proof. It is directly from Proposition 4.13 that $h_i \cdot D_{i,i+1} = h_{i+1} \cdot D_{i+1,i}$. □

Theorem 4.22. *There is a choice of construction of the d-semistable Kulikov surface $X_0(\overline{h})$, such that there is a lift of (h_i) to a nef line bundle L_0 on $X_0(\overline{h})$. Let the smoothing $X(\overline{h}) \to C$ of $X_0(\overline{h})$ keep (h_i) Cartier and then there is a nef line bundle L on $X(\overline{h})$ which when restricted to the central fiber is L_0. Moreover, on the general fiber the bundle L_t is primitive.*

Proof. The period point $\psi_{X_0(\overline{h})}$ measures the obstruction to a class being Cartier. Actually, (h_i) is Cartier on X_0 if and only if $\psi_{X_0(\overline{h})}((h_i))$ is trivial. This condition $(h_i) \in \ker(\psi_{X_0(\overline{h})})$ on the period map cut out a sublocus in the deformation space, locally parametrized by $\mathrm{Hom}(\Lambda(X_0(\overline{h})/(\mathbb{Z}\cdot(h_i)), E)$. Like in Thm 4.19, choose a d-semistable Kulikov surface whose period point lies in this locus and the first assertion follows. (By the construction of anticanonical pairs, (h_i) is nef on each component.)

For the Type II model $X(\overline{h}) \to C$ obtained by smoothing the d-semistable Kulikov surface, consider the isometry $c_t^* : \Lambda(X_0(\overline{h})) \to I^\perp/I$ defined in Prop 2.15. The class $c_1(L_t) \in I^\perp \subset H^2(X_t(\overline{h}))$ of the line bundle L_t is a lift of the image $c_t^*((h_i))$. So when (h_i) is primitive in $\Lambda(X_0(\overline{h}))$, L_t is automatically primitive.

On the other hand, when (h_i) is imprimitive, then the quotient $\Lambda(X_0(\overline{h})/(\mathbb{Z}\cdot(h_i))$ is torsion. The period point $\psi_{X_0(\overline{h})}$ belongs to the locus $\mathrm{Hom}(\Lambda(X_0(\overline{h})/(\mathbb{Z}\cdot(h_i))^{sat}, E)$ and is determined by a homomorphism $\Psi : \mathrm{Tors}(\Lambda(X_0(\overline{h})/(\mathbb{Z}\cdot(h_i)))) \to E$. When Ψ is injective, L_0 is primitive as a line bundle on $X_0(\overline{h})$, so the lift L is primitive too. Thus the second assertion holds. □

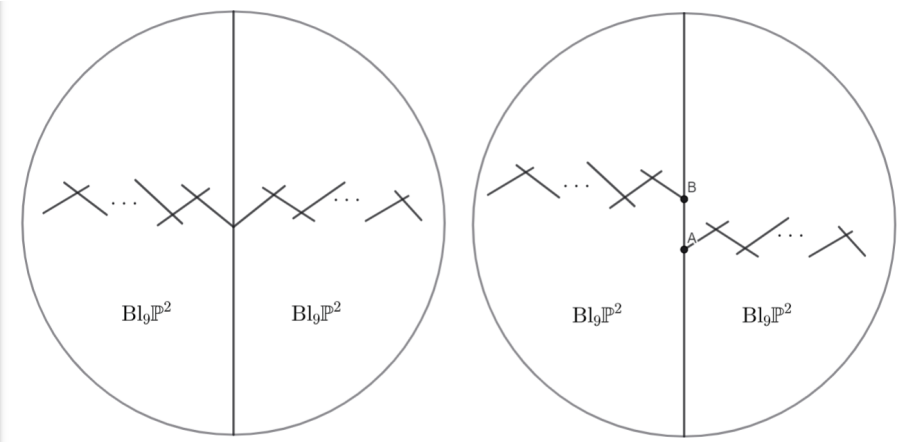


FIGURE 6. Two Type II Kulikov surfaces $X_0 = \mathrm{Bl}_9\mathbb{P}^2 \cup \mathrm{Bl}_9\mathbb{P}^2$ with the configuration of blow-ups. Left: a standard surface. Right: a non-standard surface, where the numerically Cartier divisor (e_9, e_{10}) supported on two components is mapped to $B - A \in E$ by the period map ψ_{X_0}.

Remark 4.23. A *standard surface* is a Kulikov surface with period point $\psi_{X_0} \equiv 1 \in \mathrm{Hom}(\widetilde{\Lambda}, \mathbb{C}^*)$ or $\mathrm{Hom}(\widetilde{\Lambda}, E)$. In [ABE22] the standard surface is chosen to glue the class of fibers and section, but here the second statement cannot hold for a standard surface. E.g. for the non-standard Kulikov surface in Figure 6, suppose $B - A \in E$ is k-torsion, then $k(e_9, e_{10})$ is Cartier since $\psi_{X_0}(k(e_9, e_{10})) = 1 \in E$.

The nef line bundle $L_0 \to X_0(\overline{h})$ satisfies $c_1(L_0|_{V_i}) = h_i$. Hence $L_0^2 = \sum_i h_i^2 =$ integral-affine area of $Q(\overline{h}) = \overline{h}^2$. In fact, the isometry c_t^* implies that $\Lambda(X_0(\overline{h})) \cong \mathrm{II}_{1,17}$ and we have

Theorem 4.24. *Under the isometry* $\Lambda(X_0(\overline{h})) \cong II_{1,17}$,

$$(h_i) = \overline{h} \in II_{1,17}.$$

Proof. To prove the theorem, we construct 19 simple roots in $\Lambda(X_0(\overline{h}))$ such that they generate the same root system as the α_i's which define the fundamental chamber $\mathfrak{K}$ of $\mathrm{II}_{1,17}$. Suppose there are n exceptional curves in $X_0(\overline{h})$. By Remark 4.18, $n = 16, 17$ or 18.

We choose such roots in $\bigoplus_{i=1}^r \mathrm{Pic}(V_i)$ first, using the classes in end components and classes of those exceptional curves, dependent of the type of the end components. The selection process is divided into two steps. The first step is to take 19 roots according to the following table.

TABLE 3. 19 roots in $\bigoplus_{i=1}^r \mathrm{Pic}(V_i)$ before modifying symbols

Left end component	Left end roots	Intermediate roots	Right end roots	Right end component
E-type	$l_1 - e_1 - e_2 - e_3$	$e_i - e_{i+1}, i = 1, \cdots, n-1$	$l_r + e_{n-2} + e_{n-1} + e_n$	E-type
D-type	$f_1 - e_1 - e_2, s_1 - f_1$		$f_r + e_{n-1} + e_n, s_r - f_r$	D-type

The second step is to modify the symbols of e_i in these roots: for a certain k, we change the symbol of all the classes $e_i, i > k$ in each root. When $k \neq n$, we get modified roots $e_k + e_{k+1}$, $-e_i + e_{i+1}, i > k$ and a modified right end root (maybe also a modified left end root, if $k \leq 3$. It

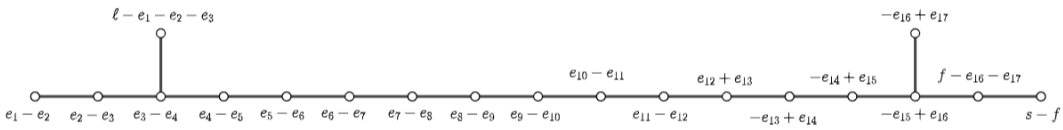


FIGURE 7. 19 roots in $\bigoplus_{i=1}^{r} \mathrm{Pic}(V_i)$ of a Kulikov surface with E-type left end and D-type right end.

turns out that such 19 roots always have the same intersections as those in the fundamental chamber $\mathfrak{K}$ of $\mathrm{II}_{1,17}$, as the Figure 7 shows, where $k = 12$. The position k of the change of symbol can be chosen at will. We always choose k such that e_{k+1} is the first exceptional curve on the right end component if there exist exceptional curves on it, or otherwise choose $k = n$, i.e. no modification of symbols.

To lift these roots into $\Lambda(X_0(\overline{h}))$, we show that there are 19 numerically Cartier classes corresponding to these root. When the classes in a root are on the same component, by our construction of anticanonical pairs (Table. 1 and 2) the intersection number of the root with the double curves must be 0. Then the root can be regarded as in $\Lambda(X_0(\overline{h}))$ directly, being zero on other components.

When classes in a root are on different components, we modify by adding the class of ruling on some intermediate component, to obtain a root in $\Lambda(X_0(\overline{h}))$. For example, suppose e_i and e_{i+1} are in components V_i and V_{i+1} respectively. Then $(0, \cdots, e_i, f_{i+1} - e_{i+1}, \cdots, 0)$ is a root in Λ, corresponding to the root $e_i - e_{i+1} \in \bigoplus_{i=1}^{r} \mathrm{Pic}(V_i)$, because $e_i.D_{i,i+1} = (f_{i+1} - e_{i+1}).D_{i+1,i} = 1$ and $(f_{i+1} - e_{i+1}).D_{i+1,i+2} = 0$. We list the choices of roots and the corresponding lifts in Table 4, when classes in a root may be on different components. The lifting does not change the intersection among the 19 roots, so we obtain the wanted root system in $\Lambda(X_0(\overline{h})) \cong \mathrm{II}_{1,17}$. Since such configuration of roots is unique up to isometry $O(\mathrm{II}_{1,17})$, we can regard these roots as the ones in the Coxeter diagram of $\mathrm{II}_{1,17}$.

Finally it remains to show that the intersection of L_0 and the 19 roots above in $\Lambda(X_0(\overline{h}))$ gives the same coefficients $(\ell_i) \in \mathbb{Z}^{19}$ as in Section 3.5. This can be done by checking for each case of components. We omit the computations here. □

Remark 4.25. Under our choice of the position k of change of symbols, suppose $X_0(\overline{h})$ is standard, then the lifted numerically Cartier divisors are (-2)-curves, the configuration of which can be visualized like in Figure 6.

Theorem 4.26. *For any class of Type II pair (I, h), we constructed a model $(X, L) \to C$ of a Type II degeneration of primitively polarized K3 surfaces of degree h^2, such that the monodromy invariant lattice is I and $c_1(L_t) = h \in H^2(X_t)$.*

Proof. Input $\overline{h} \in I^{\perp}/I \simeq \mathrm{II}_{1,17}$ to the construction, we get a Type II Kulikov model $(X(\overline{h}), L) \to C$. By Theorem 4.22 and 4.24, the class $h' := c_1(L_t) \in J^{\perp} \subset H^2(X_t(\overline{h}))$ is a primitive lift of $\overline{h}$, where $J \subset H^2(X_t(\overline{h}))$ is the rank 2 saturated isotropic sublattice of monodromy invariants. By Theorem 4.8, the class of primitive lifts is unique, so Type II pairs (I, h) and (J, h') are in the same class. □

TABLE 4. Possible roots and corresponding numerically Cartier divisor

Roots in $\bigoplus_{i=1}^{r} \mathrm{Pic}(V_i)$	Relation between curves and components	Lifted numerically Cartier divisor
$l_1 - e_1 - e_2 - e_3$	e_1, e_2, e_3 on the E-type left end V_1	$(l_1 - e_1 - e_2 - e_3, 0, \cdots)$
$l_1 - e_1 - e_2 - e_3$	$e_1, e_2 \in \mathrm{Pic}(V_1)$, $e_3 \in \mathrm{Pic}(V_2)$ for V_2 intermediate	$(l_1 - e_1 - e_2, f_2 - e_3, 0, \cdots)$
$l_1 - e_1 - e_2 + e_3$	$e_1, e_2 \in \mathrm{Pic}(V_1)$, $e_3 \in \mathrm{Pic}(V_2)$ for V_2 the right end	$(l_1 - e_1 - e_2, e_3)$
$l_1 - e_1 - e_2 - e_3$	$e_1 \in \mathrm{Pic}(V_1)$, $e_2, e_3 \in \mathrm{Pic}(V_2)$ for V_2 intermediate	$(l_1 - e_1, 2f_2 - e_2 - e_3, 0, \cdots)$
$l_1 - e_1 + e_2 + e_3$	$e_1 \in \mathrm{Pic}(V_1)$, $e_2, e_3 \in \mathrm{Pic}(V_2)$ for V_2 the right end	$(l_1 - e_1, e_2 + e_3)$
$l_1 - e_1 - e_2 - e_3$	$e_1, e_2, e_3 \in \mathrm{Pic}(V_2)$ for V_2 intermediate	$(l_1, 3f_2 - e_1 - e_2 - e_3, 0, \cdots)$
$l_1 + e_1 + e_2 + e_3$	$e_1, e_2, e_3 \in \mathrm{Pic}(V_2)$ for V_2 the right end	$(l_1, e_1 + e_2 + e_3)$
$l_1 - e_1 - e_2 - e_3$	$e_1 \in \mathrm{Pic}(V_1)$, $e_2 \in \mathrm{Pic}(V_2)$, $e_3 \in \mathrm{Pic}(V_3)$, V_2, V_3 intermediate	$(l_1 - e_1, 2f_2 - e_2, f_3 - e_3, 0, \cdots)$
$l_1 - e_1 - e_2 + e_3$	$e_1 \in \mathrm{Pic}(V_1)$, $e_2 \in \mathrm{Pic}(V_2)$, $e_3 \in \mathrm{Pic}(V_3)$, V_3 the right end	$(l_1 - e_1, 2f_2 - e_2, e_3)$
$l_1 - e_1 - e_2 - e_3$	$e_1, e_2 \in \mathrm{Pic}(V_2)$, $e_3 \in \mathrm{Pic}(V_3)$, V_3 intermediate	$(l_1, 3f_2 - e_1 - e_2, f - e_3, 0, \cdots)$
$l_1 - e_1 - e_2 + e_3$	$e_1, e_2 \in \mathrm{Pic}(V_2)$, $e_3 \in \mathrm{Pic}(V_3)$, V_3 the right end	$(l_1, 3f_2 - e_1 - e_2, e_3)$
$l_1 - e_1 - e_2 - e_3$	$e_1 \in \mathrm{Pic}(V_2)$, $e_2, e_3 \in \mathrm{Pic}(V_3)$, V_3 intermediate	$(l_1, 3f_2 - e_1, 2f_3 - e_2 - e_3, 0, \cdots)$
$l_1 - e_1 + e_2 + e_3$	$e_1 \in \mathrm{Pic}(V_2)$, $e_2, e_3 \in \mathrm{Pic}(V_3)$, V_3 the right end	$(l_1, 3f_2 - e_1, e_2 + e_3)$
$l_1 - e_1 - e_2 - e_3$	$e_1 \in \mathrm{Pic}(V_2)$, $e_2 \in \mathrm{Pic}(V_3)$, $e_3 \in \mathrm{Pic}(V_4)$, V_4 intermediate	$(l_1, 3f_2 - e_1, 2f_3 - e_2, f_4 - e_3, 0, \cdots)$
$l_1 - e_1 - e_2 + e_3$	$e_1 \in \mathrm{Pic}(V_2)$, $e_2 \in \mathrm{Pic}(V_3)$, $e_3 \in \mathrm{Pic}(V_4)$, V_4 the right end	$(l_1, 3f_2 - e_1, 2f_3 - e_2, e_3)$
$s_1 - f_1$	s_1, f_1 classes of rulings on the D-type left end V_1	$(s_1 - f_1, 0, \cdots)$
$f_1 - e_1 - e_2$	e_1, e_2 on the D-type left end V_1	$(f_1 - e_1 - e_2, 0, \cdots)$
$f_1 - e_1 - e_2$	$e_1 \in \mathrm{Pic}(V_1)$, $e_2 \in \mathrm{Pic}(V_2)$ for V_2 intermediate	$(f_1 - e_1, f_2 - e_2, 0, \cdots)$
$f_1 - e_1 + e_2$	$e_1 \in \mathrm{Pic}(V_1)$, $e_2 \in \mathrm{Pic}(V_2)$ for V_2 the right end	$(f_1 - e_1, e_2)$
$f_1 - e_1 - e_2$	$e_1, e_2 \in \mathrm{Pic}(V_2)$ for V_2 intermediate	$(f_1, 2f_2 - e_1 - e_2, 0, \cdots)$
$f_1 + e_1 + e_2$	$e_1, e_2 \in \mathrm{Pic}(V_2)$ for V_2 the right end	$(f_1, e_1 + e_2)$
$f_1 - e_1 - e_2$	$e_1 \in \mathrm{Pic}(V_2)$, $e_2 \in \mathrm{Pic}(V_3)$, V_2, V_3 intermediate	$(f_1, 2f_2 - e_1, f_3 - e_2, 0, \cdots)$
$f_1 - e_1 + e_2$	$e_1 \in \mathrm{Pic}(V_2)$, $e_2 \in \mathrm{Pic}(V_3)$, V_3 the right end	$(f_1, 2f_2 - e_1, e_2)$
$e_i - e_{i+1}$	exceptional curves on the same component	$(\cdots, 0, e_i - e_{i+1}, 0, \cdots)$
$e_i - e_{i+1}$	exceptional curves on adjacent components	$(\cdots, 0, e_i, f - e_{i+1}, 0, \cdots)$
$e_i + e_{i+1}$	e_{i+1} the first exceptional curve on the right end component	$(\cdots, 0, e_i, e_{i+1})$
$-e_i + e_{i+1}$	exceptional curves on the right end component	$(\cdots, 0, -e_i + e_{i+1})$
$l_r - e_{n-2} - e_{n-1} - e_n$	e_{n-2}, e_{n-1}, e_n on the E-type right end V_r	$(\cdots, 0, l_r - e_{n-2} - e_{n-1} - e_n)$
$l_r + e_{n-2} - e_{n-1} - e_n$	$e_{n-1}, e_n \in \mathrm{Pic}(V_r)$, $e_{n-2} \in \mathrm{Pic}(V_{r-1})$	$(\cdots, 0, e_{n-2}, l_r - e_{n-1} - e_n)$
$l_r + e_{n-2} + e_{n-1} - e_n$	$e_n \in \mathrm{Pic}(V_r)$, $e_{n-2}, e_{n-1} \in \mathrm{Pic}(V_{r-1})$	$(\cdots, 0, e_{n-2} + e_{n-1}, l_r - e_n)$
$l_r + e_{n-2} + e_{n-1} + e_n$	$e_{n-2}, e_{n-1}, e_n \in \mathrm{Pic}(V_{r-1})$	$(\cdots, 0, e_{n-2} + e_{n-1} + e_n, l_r)$
$l_r + e_{n-2} + e_{n-1} - e_n$	$e_n \in \mathrm{Pic}(V_r)$, $e_{n-1} \in \mathrm{Pic}(V_{r-1})$, $e_{n-2} \in \mathrm{Pic}(V_{r-2})$	$(\cdots, 0, e_{n-2}, f_{r-1} + e_{n-1}, l_r - e_n)$
$l_r + e_{n-2} + e_{n-1} + e_n$	$e_{n-1}, e_n \in \mathrm{Pic}(V_{r-1})$, $e_{n-2} \in \mathrm{Pic}(V_{r-2})$	$(\cdots, 0, e_{n-2}, f_{r-1} + e_{n-1} + e_n, l_r)$
$l_r + e_{n-2} + e_{n-1} + e_n$	$e_n \in \mathrm{Pic}(V_{r-1})$, $e_{n-1}, e_{n-2} \in \mathrm{Pic}(V_{r-2})$	$(\cdots, 0, e_{n-2} + e_{n-1}, 2f_{r-1} + e_n, l_r)$
$l_r + e_{n-2} + e_{n-1} + e_n$	$e_n \in \mathrm{Pic}(V_{r-1})$, $e_{n-1} \in \mathrm{Pic}(V_{r-2})$, $e_{n-2} \in \mathrm{Pic}(V_{r-3})$	$(\cdots, 0, e_{n-2}, f_{r-2} + e_{n-1}, 2f_{r-1} + e_n, l_r)$
$s_r - f_r$	s_r, f_r classes of rulings on the D-type right end V_r	$(\cdots, 0, s_r - f_r)$

$f_r - e_{n-1} - e_n$	e_{n-1}, e_n on the D-type right end V_r	$(\cdots, 0, f_r - e_{n-1} - e_n)$
$f_r + e_{n-1} - e_n$	$e_{n-1} \in \mathrm{Pic}(V_{r-1}), e_n \in \mathrm{Pic}(V_r)$	$(\cdots, 0, e_{n-1}, f_r - e_n)$
$f_r + e_{n-1} + e_n$	$e_{n-1}, e_n \in \mathrm{Pic}(V_{r-1})$	$(\cdots, 0, e_{n-1} + e_n, f_r)$
$f_r + e_{n-1} + e_n$	$e_{n-1} \in \mathrm{Pic}(V_{r-2}), e_n \in \mathrm{Pic}(V_{r-1})$	$(\cdots, 0, e_{n-1}, f_{r-1} + e_n, f_r)$

Remark 4.27. A class of vector $\overline{h} \in \mathrm{II}_{1,17}$ can have two representatives in the chamber $\mathfrak{K}$, with left-right inverse intersections with the roots. Generally the Kulikov surfaces obtained from the two reprensentatives are not isomorphic, but by [FM83b] the Kulikov models are related by Atiyah flops and topologically trivial deformations.

4.3. **Examples.** Here we give three examples of Kulikov surfaces $X_0(\overline{h})$, see Figure 8). The notations a_i, α_i, ℓ_i are from Section 3.5. The notations for classes of curves follow from last section.

Example 4.28. The lattice polygon of the first example is a single triangle determined by the point $(0.5, 2)$ (i.e. singularity X_4'), corresponding to the lattice vector $\overline{h} \in \mathrm{II}_{1,17}$ characterized by $\ell_2 = c_L/2 = \frac{1}{2}$, $\ell_{19} = a_{18} = 2$ and $\ell_i = 0$ $(i \neq 2, 19)$ (cf. Section 3.5).

The left end component associated by the triangle is $(V_1 = \mathbb{P}^1 \times \mathbb{P}^1, D_{12} \in |2s_1 + 2f_1|)$, with the polarization $h_1 = s_1 + f_1$ on it. Since there is only one strip, the right end component is $(V_2 = \mathrm{Bl}_{16}(\mathbb{P}^1 \times \mathbb{P}^1), D_{21} \in |2s_2 + 2f_2 - e_1 - \cdots - e_{16}|)$ with polarization $h_2 = 2f_2$. In particular, all 16 blow ups on (the anticanonical elliptic curve of) V_2 are additional blow-ups.

Thus the 19 roots in the Coxeter diagram of $\Lambda(X_0(\overline{h}))$ (easy to see they are numerically Cartier) are:

$$\begin{cases} \alpha_1 = (0, -e_1 + e_2), \alpha_2 = (s_1 - f_1, 0), \alpha_3 = (f_1, e_1 + e_2), \\ \alpha_k = (0, -e_{k-2} + e_{k-1}) \text{ for } k = 4, \cdots, 16, \\ \alpha_{17} = (0, f_2 - e_{15} - e_{16}), \alpha_{18} = (0, s_2 - f_2), \alpha_{19} = (0, -e_{15} + e_{16}). \end{cases}$$

And one can check the vector of polarization $\overline{h} = (s_1 + f_1, 2f_2)$ has intersections $a_3 = 1$, $a_{18} = 2$ and $a_i = 0$ for $i \neq 3, 18$. Particularly, it induces the Type II degeneration of degree 2 K3 surfaces of combinatorial type $D_{16}A_1$, i.e. the type of root system of $\overline{h}^{\perp} \subset \mathrm{II}_{1,17}$.

Example 4.29. The second lattice polygon consists of two trapezoids corresponding to two singularities of type Y_2Y_2', determined by 3 points $(0, 0.5)$, $(0.5, 2.5)$, and $(1, 0.5)$. So $\overline{h}$ is given by

$$\begin{cases} \ell_1 = b_L/2 = \frac{1}{2}, \ell_2 = -c_L/2 = \frac{1}{2}, \\ \ell_i = 0 \text{ for } i = 3, \cdots, 17, \\ \ell_{18} = -c_R/2 = \frac{1}{2}, \ell_{19} = b_R/2 = \frac{1}{2}. \end{cases}$$

Then, we have left end component $(V_1 = \mathrm{Bl}_1(\mathbb{P}^1 \times \mathbb{P}^1), D_{12} \in |2s_1 + 2f_1 - e_1|)$ with $h_1 = s_1 + 2f_1 - e_1$, and right one $(V_2 = \mathrm{Bl}_{15}(\mathbb{P}^1 \times \mathbb{P}^1), D_{21} \in |2s_2 + 2f_2 - e_2 - \cdots - e_{16}|)$ with $h_2 = s_2 + 2f_2 - e_{16}$. The blow-ups giving rise to e_2 to e_{15} are additional blow-ups on V_2.

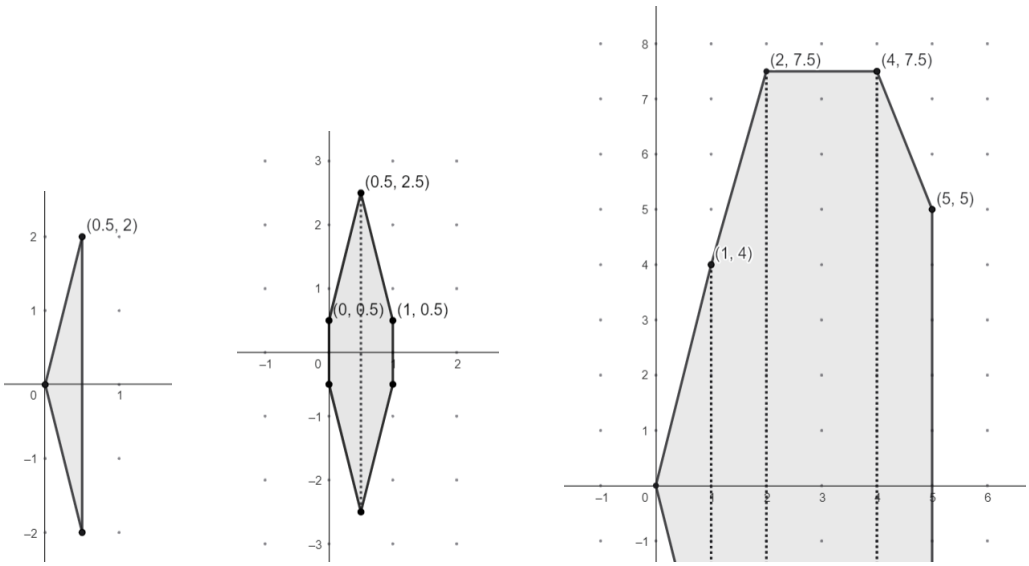


FIGURE 8. Lattice polygons of examples.

Thus the 19 roots in the Coxeter diagram are:

$$\begin{cases} \alpha_1 = (e_1, e_2), \alpha_2 = (s_1 - f_1, 0), \alpha_3 = (f_1 - e_1, e_2), \\ \alpha_k = (0, -e_{k-2} + e_{k-1}) \text{ for } k = 4, \cdots, 16, \\ \alpha_{17} = (0, f_2 - e_{15} - e_{16}), \alpha_{18} = (0, s_2 - f_2), \alpha_{19} = (0, -e_{15} + e_{16}). \end{cases}$$

One can check the intersections with $\overline{h} = (s_1 + 2f_1 - e_1, s_2 + 2f_2 - e_{16})$: the nonzero ones are $a_1 = a_2 = a_{18} = a_{19} = 1$. Particularly, it will induce a Type II degeneration of K3 surfaces of degree 6.

Example 4.30. The third one has 4 strips from the picture, which will in turn give 4 components. $\overline{h}$ is given by

$$\begin{cases} \ell_1 = -b_L/3 = 0, \ell_2 = a_2 = 1, \ell_3 = a_3 = 1, \\ \ell_{10} = a_{10} = 2, \ell_{15} = a_{15} = 1, \ell_{19} = a_{18} = 5, \\ \ell_i = 0 \text{ for other } i. \end{cases}$$

The left end component is $(V_1 = \mathrm{Bl}_1\mathbb{P}^2, D_{12} \in |3l - e_1|)$ with $h_1 = 3l - e_1$.

The second component is a $\mathbb{P}^1$ fibration over an elliptic curve with two sections $D_{21}^2 = s_{2,0}^2 = -8$ and $D_{23}^2 = (s_{2,\infty} - e_2)^2 = 8 - 1 = 7$, with $h_2 = 8f_2 + s_{2,\infty} - e_2$.

The third component is similarly given by $D_{32}^2 = s_{3,0}^2 = -7$ and $D_{34}^2 = (s_{3,\infty} - e_3 - \cdots - e_9)^2 = 7 - 7 = 0$, with $h_3 = 15f_3 + 2(s_{3,\infty} - e_3 - \cdots - e_9)$.

The right end component is $(V_4 = \mathrm{Bl}_8(\mathbb{P}^1 \times \mathbb{P}^1), D_{43} \in |2s_4 + 2f_4 - e_{10} - \cdots - e_{17}|)$ with $h_4 = 2s_4 + 7f_4 - e_{15} - e_{16} - e_{17}$. The blow-ups giving rise to e_{10} to e_{14} are additional blow-ups.

The 19 roots in the Coxeter diagram are:

$$\begin{cases} \alpha_1 = (l - e_1, 2f_2 - e_2, f_3 - e_3, 0), \\ \alpha_2 = (e_1, f_2 - e_2, 0, 0), \\ \alpha_3 = (0, e_2, f_3 - e_3, 0), \\ \alpha_k = (0, 0, e_{k-1} - e_k, 0) \text{ for } k = 4, \cdots, 9, \\ \alpha_{10} = (0, 0, e_9, e_{10}), \\ \alpha_k = (0, 0, 0, -e_{k-1} + e_k) \text{ for } k = 11, \cdots, 16, \\ \alpha_{17} = (0, 0, 0, f_4 - e_{16} - e_{17}), \alpha_{18} = (0, 0, 0, s_4 - f_4), \alpha_{19} = (0, 0, 0, -e_{16} + e_{17}). \end{cases}$$

The nonzero intersections with $\overline{h} = (h_i)$: $a_1 = 2 + 1 + 0 + 0 = 3$, $a_2 = 1 + 0 + 0 + 0 = 1$, $a_3 = 0 + 1 + 0 + 0 = 1$, $a_{10} = 0 + 0 + 2 + 0 = 2$, $a_{15} = 0 + 0 + 0 + 1 = 1$, and $a_{18} = 0 + 0 + 0 + 5 = 5$. In particular, it will induce a degeneration of K3 surfaces of degree $8 + 23 + 60 + 25 = 116$.

Remark 4.31. See in these examples how the roots are transformed from the ones in $\bigoplus_{i=1}^{r} \mathrm{Pic}(V_i)$, comparing with the proof of Thm 4.24. It is also easy to check Prop. 4.13 in these examples.

4.4. **Generalizations and Conjectures.** Reviewing what has been done, it is natural to regard the lattice polygon $Q(\overline{h})$ as a kind of combinatorial structure over the dual complex of the Kulikov surface $X_0(\overline{h})$: the segment is the intersection of $Q(\overline{h})$ and the x-axis. So we answered the Question 1.1 at the beginning.

In fact, the information we input, a lattice vector $\overline{h} \in \mathrm{II}_{1,17}$, is about the polarizing divisor. Therefore, the part of construction for the Kulikov surface is more like a choice, allowing the wanted polarizing divisor to be simply represented on it. So the combinatorial structure should be regarded as characterizing the polarizing divisor. This inspires us to compare it with the definition 3.30 of integral-affine divisor in the Type III case, and ask: is there some kind of relationship between these two combinatorial structures, say lattice polygons and weighted graphs, within a larger framework of integral affine geometry?

In fact, such affine geometry has already been studied by Gross-Siebert [GS12]. In this section, we attempt to present it under Engel's framework, namely pseudo fans and IAS^2.

Definition 4.32. On an integral affine surface S, a $\mathbb{Z}$-linear function f is defined on its nonsingular locus, of the form $f(\mathbf{x}) = a_1 x_1 + a_2 x_2 + b$ with $a_i \in \mathbb{Z}$ on any affine chart U with local coordinates x_i. Define $\mathbb{Z}\mathcal{L}$ to be the sheaf of locally $\mathbb{Z}$-linear functions and $\mathbb{Z}\mathcal{PL}$ to be the sheaf of continuous, piecewise $\mathbb{Z}$-linear functions.

Then $\mathbb{Z}\mathcal{L}$ is a subsheaf of $\mathbb{Z}\mathcal{PL}$. Suppose S is a non-singular integral affine surface, e.g., polytopes, $\mathbb{R}^2$ or $\mathbb{T}^2$.

Definition 4.33. The *bending locus* of a function $f \in H^0(S, \mathbb{Z}\mathcal{PL})$ is a weighted graph $\mathrm{Bend}(f)$ on S, whose edges are the locus where f is not linear, and the weight $w(e) \in \mathbb{Z}$ on an edge e of $\mathrm{Bend}(f)$ is the index of $\mathbf{a} - \mathbf{a}'$ where $\mathbf{a}$, $\mathbf{a}' \in \mathbb{Z}^2$ give the affine part of f on the two domains separated by e.

Definition 4.34. For a vertex v of a weighted graph G in an open affine U, let e_i be the edges containing v and $\mathbf{e}_i(v)$ be the primitive tangent vectors to e_i originating from v. The *balancing*

condition for v is $\sum_i w(e_i)\mathbf{e}_i(v) = 0$. We say a weighted graph G on an integral affine surface S is *balanced* if each vertex v of G satisfies the balancing condition in some affine $U \ni v$.

Proposition 4.35. *The bending locus of a function $f \in H^0(S, \mathbb{Z}\mathcal{PL})$ is balanced, the weights of whose edges are all positive if f is convex. Moreover, it determines the function up to $\mathbb{Z}$-linear functions, hence a global section $\overline{f} \in H^0(S, \mathbb{Z}\mathcal{PL}/\mathbb{Z}\mathcal{L})$.*

Remark 4.36. These convex sections appear in the Mumford degeneration of PPAVs, providing polarization on degeneration. In fact, these sections alone can completely determine a polarized Mumford degeneration by constructing what is called a *theta function*. See the recent excellent survey [EdGFS25] on this and compare the *fan construction* and *polytope construction* loc.cit with IAS^2.

Gross-Siebert introduced a duality transformation called the *discrete Legendre transformation* on polarized tropical manifolds. Under our definitions above, this transformation reduces to a much simpler, non-singular, 2-dimensional case. Briefly, it transforms (S, G), a non-singular integral affine surface with a positive balanced graph on it, into its dual $(\check{S}, \check{G})$, such that the 2-cells of $\check{G}$ correspond to vertices of G, and edges in $\check{G}$ of weight w and index k correspond to edges in G of weight k and index w.

Example 4.37. Consider a complete fan Σ as a balanced graph on $\mathbb{R}^2$, then it is transformed to its dual polytope P with the boundary ∂P.

This transformation aims at studying mirror symmetry by tropical geometry. To study the mirror symmetry of K3 surfaces, we should also define the discrete Legendre transformation of IAS^2. An integral affine divisor R_{IA} should play the role of bending locus, and we want to construct a dual $(\check{B}, \check{R}_{\mathrm{IA}})$ of (B, R_{IA}).

Naturally, we wanna maintain the correspondence between 2-cells and vertices, and between edges. Then, we consider how to work on integral affine singularities.

Example 4.38. Review the two Symington surgeries on polytopes (cf. Def. 3.20) and the Example 4.37, we claim that if an I_1-singularity on B results from a internal blow-up of size n on an edge E of a polytope P, then the corresponding cone point p of P in $\check{B}$ is an I_1 singularity, resulting from a node smoothing of size n along the edge E' pointing away from p. Moreover, as an internal blow-up decreases the index of E by n, the weight of E' will also decrease by n.

Since we can factorize the singularitied on an IAS^2 B into I_1 singularities, one feasible strategy is to consider all types of I_1 singularities and then just glue the polytopes together. But the most important thing is to make this transformation compatible with the mirror symmetry of K3 surfaces. Concretely, it should have the following property:

Conjecture 4.39. *Let B be a triangulated IAS^2. Let $(X, L) \to C$ be a Type III Kulikov degeneration of polarized K3 surfaces such that $\Gamma(X_0) = B$ and L defines the integral affine divisor R_{IA} on B. The monodromy invariant and the polarization of the degeneration have images $\lambda_1, \lambda_2 \in \delta^\perp/\delta \cong II_{2,18}$.*

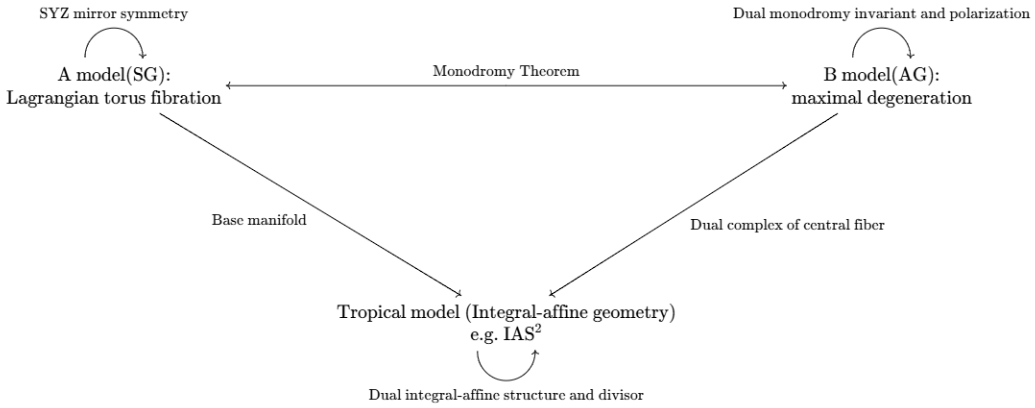


FIGURE 9. Three interrelated models with mirror symmetry, and more related theories and constructions to be studied.

Suppose $(\check{B}, \check{R}_{IA})$ *is the discrete Legendre transformation of* (B, R_{IA})*. Then, we can construct a mirror Kulikov model* $(\check{X}, \check{L}) \to C$ *such that* $(\check{\lambda}_1, \check{\lambda}_2) = \phi(\lambda_2, \lambda_1)$ *with* $\phi \in O(II_{2,18})$*,* $\Gamma(\check{X}_0) = \check{B}$ *and* $\check{L}$ *defines the integral affine divisor* $\check{R}_{IA}$*.*

Moreover, if B *is generic, then by Theorem 3.26, it gives a mirror Lagrangian torus fibration* $(\check{S}, \check{\omega}) \to \check{B}$ *of* $(S, \omega) \to B$*.*

Now we see how the construction of Type II degeneration is related to Type III of elliptic K3s. In the construction of degeneration $(X(\lambda), L) \to C$ of elliptic K3 surfaces, if we remove the horizontal edges of R_{IA} on $B(\lambda)$ and corresponding sections on components of $X_0(\lambda)$, we will obtain a divisor consisting of singular elliptic curves on $X_0(\lambda)$ being the limit of singular fibers in the degeneration. The vertical edges left in R_{IA} then separate $B(\lambda)$ into circular strips. Apply the discrete Legendre transformation, each strip becomes a vertex and each vertical edge of length n and weight k becomes an edge connecting two vertices of length k and weight n. This resulting segment $\check{B}(\lambda)$ is the dual complex of a Type II Kulikov Kulikov surface and the weights give an integral affine divisor on it.

Therefore, our construction is a limiting example of Conjecture 4.39 when one of the vectors λ_i is isotropic.

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